Diagonal 8.8 mm (Type 1/1.8) CMOS solid-state Image Sensor with Square Pixel for Monochrome Cameras

IMX568-AAMJ-C

Pregius S

1. Description

The IMX568-AAMJ-C is a diagonal 8.8 mm (Type 1/1.8) CMOS active pixel type solid-state image sensor with a square pixel array and 5.10 M effective pixels. This chip features a global shutter with variable charge-integration time. This chip operates with analog 3.3 V, 2.9V, digital 1.1 V, and interface 1.8 V quadruple power supply. High sensitivity and low dark current characteristics are achieved.
(Applications: FA cameras, ITS cameras)

2. Features

- ◆ CMOS active pixel type dots
- ◆ Built-in timing adjustment circuit, H/V driver and serial communication circuit
- ◆ Global shutter function
- ◆ Input frequency 37.125 MHz / 74.25 MHz / 54 MHz
- ◆ Number of recommended recording pixels: 2448 (H) × 2048 (V) approx. 5.01 M pixels
- ◆ Readout mode
  All-pixel scan mode
  Vertical / Horizontal 1/2 Subsampling mode
  2 × 2 FD binning mode
  ROI mode
  Vertical / Horizontal ‐ Normal / Inverted readout mode
- ◆ Readout rate
  Maximum frame rate in

  All-pixel scan mode: 8 bit 96.2 frame/s, 10 bit 79.1 frame/s, 12 bit 67.2 frame/s
  (*) At high frame rates, control so as not to exceed Tj = +100 °C
- ◆ Variable-speed shutter function (resolution 1 H units)
- ◆ Pulse Output Function

  The monitor output for Exposure period (TOUT0)
  Programmable pulse output (TOUT1 and TOUT2)
- ◆ 8-bit / 10-bit / 12-bit A/D converter
- ◆ CDS / PGA function
  0 dB to 24 dB: Analog Gain (0.1 dB step)
  24.1 dB to 48 dB: Analog Gain: 24 dB + Digital Gain: 0.1 dB to 24 dB (0.1 dB step)
- ◆ I/O interface
  CSI-2 serial data output (2 Lane / 4 Lane)
- ◆ Recommended lens F number: 2.8 or more (Close side)

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3. Device Structure

- ◆ CMOS image sensor
- ◆ Image size

  Diagonal 8.8 mm (Type 1/1.8) Approx. 5.10 M pixels
- ◆ Total number of pixels

  2472 (H) × 2128 (V) Approx. 5.26 M pixels
- ◆ Number of effective pixels

  2472 (H) × 2064 (V) Approx. 5.10 M pixels
- ◆ Number of active pixels

  2472 (H) × 2064 (V) Approx. 5.10 M pixels
- ◆ Number of recommended recording pixels

  2448 (H) × 2048 (V) Approx. 5.01 M pixels
- ◆ Unit cell size

  2.74 µm (H) × 2.74 µm (V)
- ◆ Optical black

  Horizontal (H) direction: Front 0 pixels, rear 0 pixels
  Vertical (V) direction: Front 64 pixels, rear 0 pixels
- ◆ Substrate material

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4. Absolute Maximum Ratings

Item	Symbol	Rating			Unit	Remarks
Supply voltage (Analog 3.3 V)	$AV_{DD}1$	-0.3	to	4.65	V	
Supply voltage (Analog 2.9 V)	$AV_{DD}2$	-0.3	to	4.65	V	
Supply voltage (Interface 1.8 V)	OV_{DD}	-0.3	to	4.65	V	
Supply voltage (Digital 1.1 V)	DV_{DD}	-0.3	to	1.65	V	
Input voltage	VI	-0.3	to	OV_{DD} +0.3	V	Do not exceed 4.65V
Output voltage	VO	-0.3	to	OV_{DD} +0.3	V	Do not exceed 4.65V
Operating temperature	Topr	-30	to	+75	˚C	High temperature side: Tj = +100°C *1
Storage temperature	Tstg	-40	to	+85	˚C	

*1 Please control not to exceed "Tj = + 100 °C" using the thermometer function built in the image sensor.
For the thermometer function, refer to the application note of "Thermometer".

5. Recommended Operating Conditions

Item	Symbol	Min.	Typ.	Max.	Unit
Supply voltage (Analog 3.3 V)	$AV_{DD}1$	3.15	3.3	3.45	V
Supply voltage (Analog 2.9 V)	$AV_{DD}2$	2.75	2.9	3.05	V
Supply voltage (Interface 1.8 V)	OV_{DD}	1.7	1.8	1.9	V
Supply voltage (Digital 1.1 V)	DV_{DD}	1.0	1.1	1.2	V
Performance guarantee temperature	Tspec	-10	–	+60	˚C

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CONTENTS

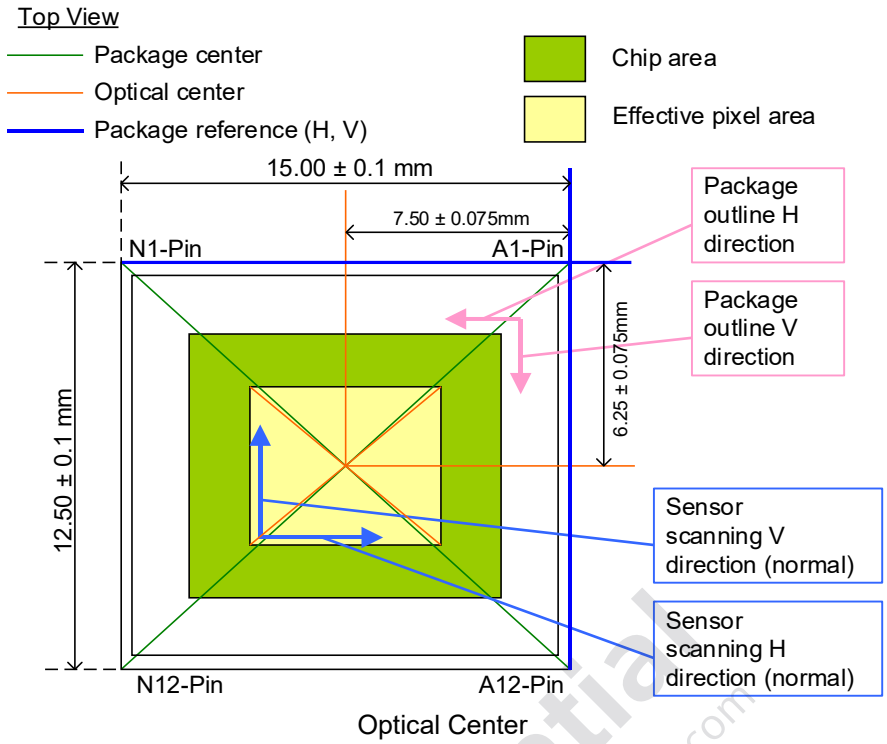
1. Description ... 1
2. Features ... 1
3. Device Structure ... 2
4. Absolute Maximum Ratings ... 3
5. Recommended Operating Conditions ... 3
6. USE RESTRICTION NOTICE ... 4
7. Chip Center and Optical Center ... 7
8. Pixel Arrangement ... 8
9. Block Diagram and Pin Configuration ... 9
10. Pin Description ... 11
11. Electrical Characteristics ... 14
    11.1. DC Characteristics ... 14
    11.2. Power Consumption ... 14
    11.3. AC Characteristics ... 15
        11.3.1. Master Clock (INCK) ... 15
        11.3.2. System Clear (XCLR) ... 16
        11.3.3. XVS / XHS Input Characteristics in Slave Mode (XMASTER = High) ... 17
        11.3.4. XTRIG Input Characteristics in Slave Mode (XMASTER = High) only ... 18
        11.3.5. Serial Communication ... 19
12. I/O Equivalent Circuit Diagram ... 21
13. Spectral Sensitivity Characteristics ... 22
14. Setting Registers Using Serial Communication ... 23
    14.1. Description of Setting Registers (4-wire) ... 23
    14.2. Register Communication Timing (4-wire) ... 24
    14.3. Register Write and Read (4-wire) ... 25
    14.4. Description of Setting Registers (I^2C) ... 26
    14.5. Register Communication Timing (I^2C) ... 27
    14.6. I^2C Communication Protocol ... 28
    14.7. I^2C Serial Communication Read/Write Operation ... 29
        14.7.1. Single Read from Random Location ... 29
        14.7.2. Single Read from Current Location ... 29
        14.7.3. Sequential Read Starting from Random Location ... 30
        14.7.4. Sequential Read Starting from Current Location ... 30
        14.7.5. Single Write to Random Location ... 31
        14.7.6. Sequential Write Starting from Random Location ... 31
15. Readout Drive Modes ... 32
16. Description of Various Functions ... 35
    16.1. Standby mode ... 35
    16.2. Slave Mode and Master Mode ... 36
    16.3. Gain Adjustment Function ... 38
    16.4. Black Level Adjustment Function ... 39
    16.5. Horizontal / Vertical Normal Operation and Inverted Operation ... 40
    16.6. Shutter and Exposure Time Settings ... 41
        16.6.1. Global Shutter (Normal Mode) Operation ... 42
        16.6.2. Global Shutter (Sequential Trigger Mode) Operation ... 45
        16.6.3. Global Shutter (Fast Trigger Mode) Operation ... 47
        16.6.4. Mode Transitions of Global Shutter Operation ... 49
        16.6.5. Pulse Output Function ... 50
    16.7. Signal Output ... 53
        16.7.1. Output Pin Settings ... 53
        16.7.2. MIPI Transmitter ... 56
        16.7.3. Output Pin Bit Width Selection ... 57
        16.7.4. Output Signal Range ... 57
    16.8. Global Timing Register Setting ... 58
    16.9. Register Hold Setting ... 59
    16.10. Mode Transition ... 60
    16.11. Other Function ... 61
    16.12. Extension Function ... 61
17. Power-on and Power-off Sequence ... 62

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17.1. Power-on sequence 62
17.2. Slew Rate Limitation of Power-on Sequence 63
17.3. Power-off Sequence 64
18. Sensor Setting Flow 65
18.1. Setting Flow in Sensor Slave Mode 65
18.2. Setting Flow in Sensor Master Mode 66
19. Peripheral Circuit 67
19.1. Analog Power Pins 67
19.2. Digital Power Pins 68
19.3. Analog Other Pins 69
19.4. Digital I/O Pins 70
19.5. Output pins 71
20. Marking 72
21. Notes On Handling 73
22. Package Outline 75
23. List of Trademark Logos and Definition Statements 76

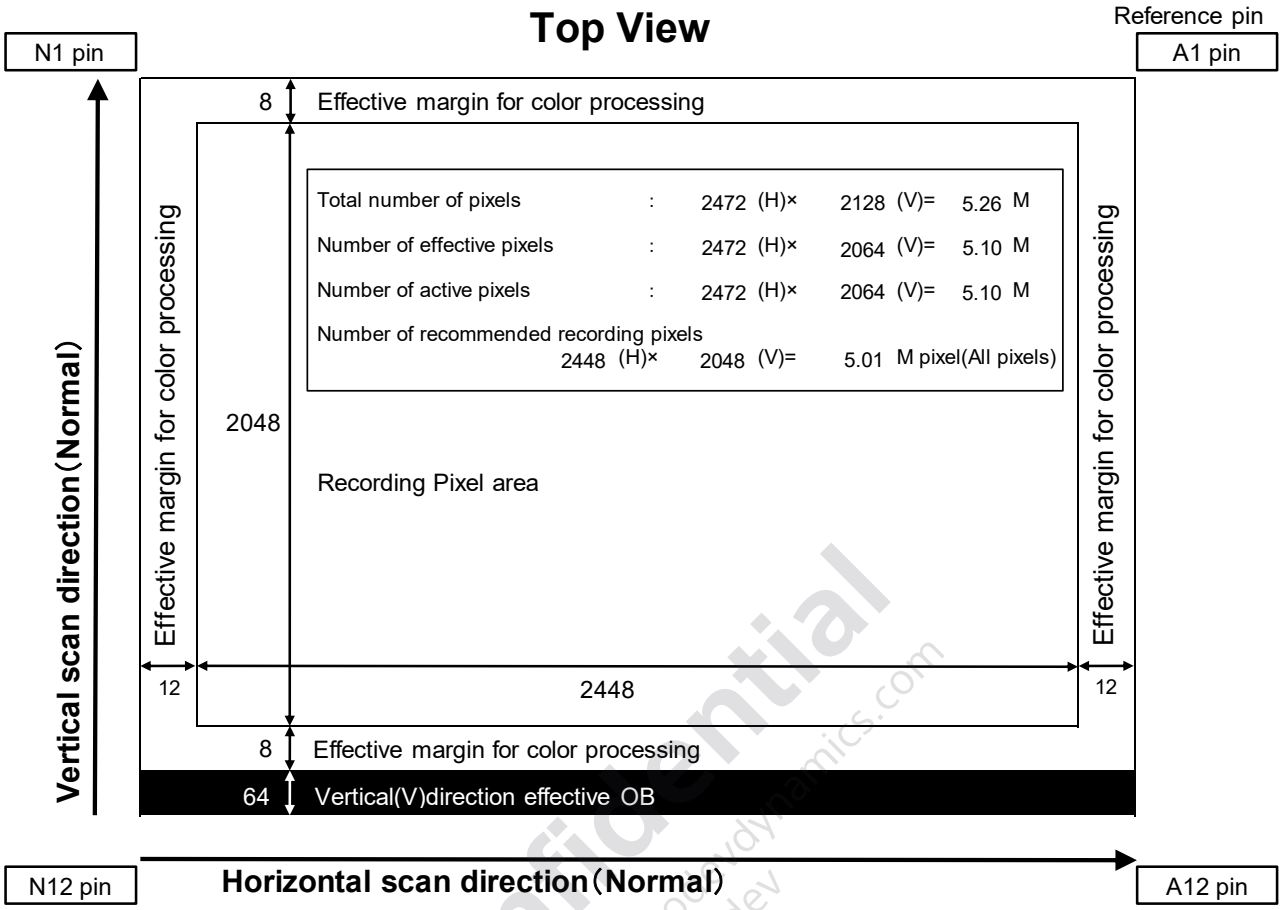
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8. Pixel Arrangement



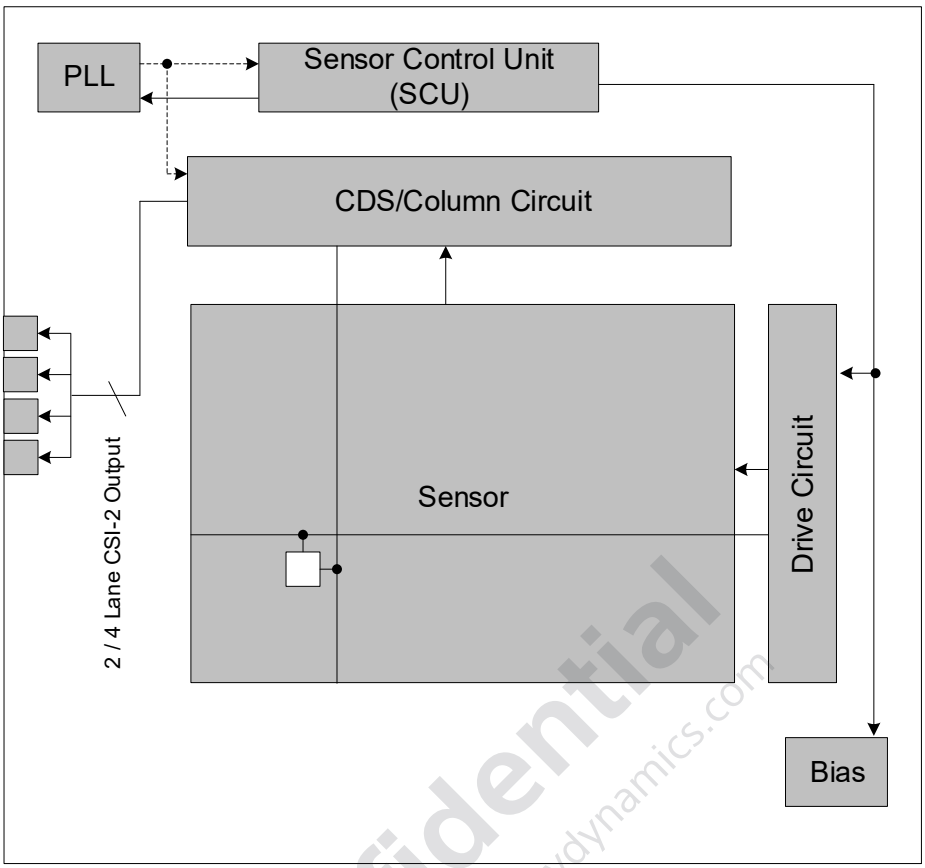
* Reference pin number is consecutive numbering of package pin array.
See the Pin Configuration for the number of each pin.

Pixel Arrangement

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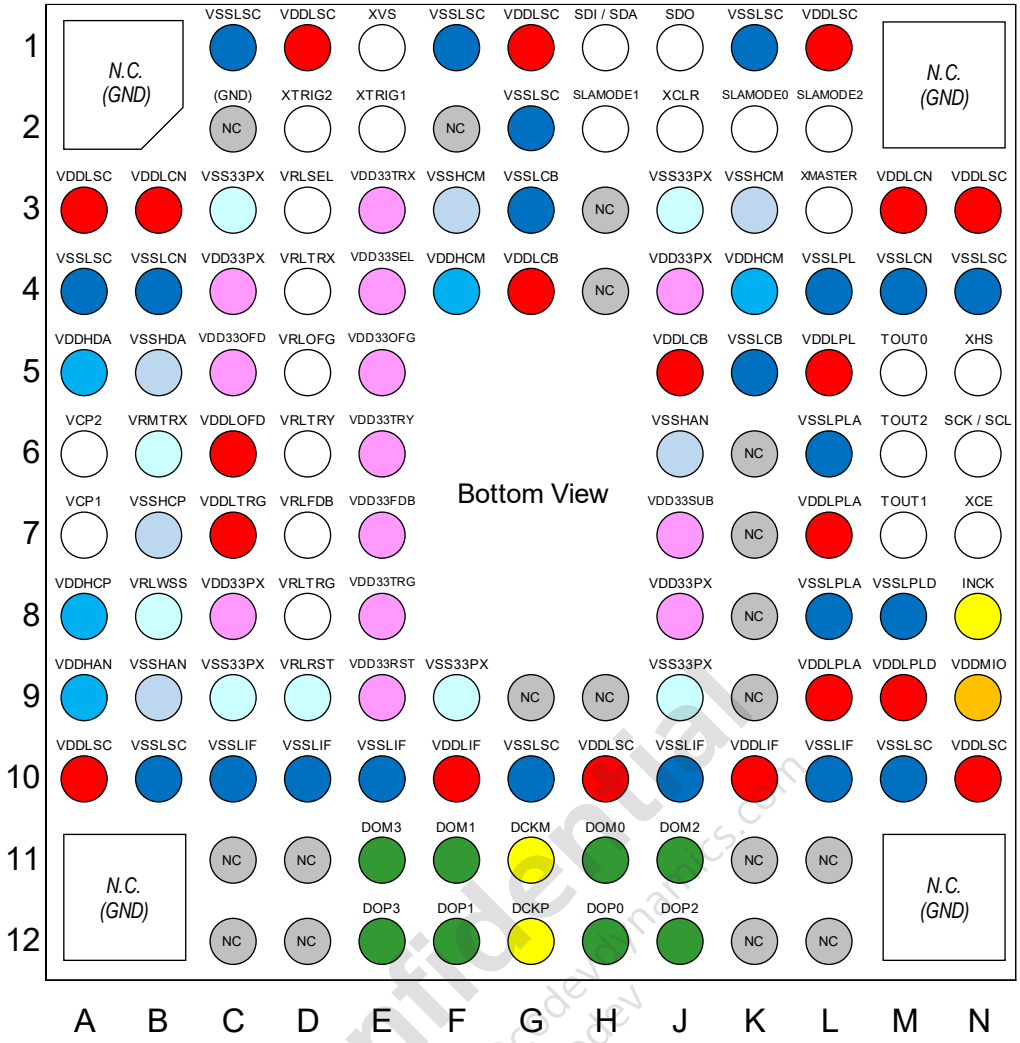
9. Block Diagram and Pin Configuration

(Top View)



Block Diagram

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*The N.C. pin with (GND) can be connected to GND.

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10. Pin Description

No.	Pin No.	I/O	Analog / Digital	Symbol	Description
1	A1	–	–	N.C.	GND connection is allowed
2	A3	Power	D	VDDLSC	1.1 V power supply
3	A4	GND	D	VSSLSC	1.1 V GND
4	A5	Power	A	VDDHDA	2.9 V power supply
5	A6	O	A	VCP2	Connect to VRLOFG, VRLSEL, VRLFDB, VRLTRX, VRLTRY (Connect 4.7 µF × 2 to GND)
6	A7	O	A	VCP1	Connect to VRLTRG (Connect 4.7 µF × 2 to GND)
7	A8	Power	A	VDDHCP	2.9 V power supply
8	A9	Power	A	VDDHAN	2.9 V power supply
9	A10	Power	D	VDDLSC	1.1 V power supply
10	A12	–	–	N.C.	GND connection is allowed
11	B3	Power	D	VDDLCN	1.1 V power supply
12	B4	GND	D	VSSLCN	1.1 V GND
13	B5	GND	A	VSSHDA	2.9 V GND
14	B6	GND	A	VRMTRX	3.3 V GND
15	B7	GND	A	VSSHCP	2.9 V GND
16	B8	GND	A	VRLWSS	3.3 V GND
17	B9	GND	A	VSSHAN	2.9 V GND
18	B10	GND	D	VSSLSC	1.1 V GND
19	C1	GND	D	VSSLSC	1.1 V GND
20	C2	–	–	N.C.	GND connection is allowed
21	C3	GND	A	VSS33PX	3.3 V GND
22	C4	Power	A	VDD33PX	3.3 V power supply
23	C5	Power	A	VDD33OFD	3.3 V power supply
24	C6	Power	A	VDDLOFD	1.1 V power supply
25	C7	Power	A	VDDLTRG	1.1 V power supply
26	C8	Power	A	VDD33PX	3.3 V power supply
27	C9	GND	A	VSS33PX	3.3 V GND
28	C10	GND	D	VSSLIF	1.1 V GND
29	C11	–	–	N.C.	–
30	C12	–	–	N.C.	–
31	D1	Power	D	VDDLSC	1.1 V power supply
32	D2	I	D	XTRIG2	Trigger input 2
33	D3	I	A	VRLSEL	Connect to VCP2
34	D4	I	A	VRLTRX	Connect to VCP2
35	D5	I	A	VRLOFG	Connect to VCP2
36	D6	I	A	VRLTRY	Connect to VCP2
37	D7	I	A	VRLFDB	Connect to VCP2
38	D8	I	A	VRLTRG	Connect to VCP1
39	D9	GND	A	VRLRST	3.3 V GND
40	D10	GND	D	VSSLIF	1.1 V GND
41	D11	–	–	N.C.	–
42	D12	–	–	N.C.	–
43	E1	I/O	D	XVS	Vertical sync signal
44	E2	I	D	XTRIG1	Trigger input 1
45	E3	Power	A	VDD33TRX	3.3 V power supply
46	E4	Power	A	VDD33SEL	3.3 V power supply
47	E5	Power	A	VDD33OFG	3.3 V power supply
48	E6	Power	A	VDD33TRY	3.3 V power supply
49	E7	Power	A	VDD33FDB	3.3 V power supply
50	E8	Power	A	VDD33TRG	3.3 V power supply
51	E9	Power	A	VDD33RST	3.3 V power supply
52	E10	GND	D	VSSLIF	1.1 V GND
53	E11	O	D	DOM3	CSI-2 IF output (Data)
54	E12	O	D	DOP3	CSI-2 IF output (Data)

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No.	Pin No.	I/O	Analog / Digital	Symbol	Description
55	F1	GND	D	VSSLSC	1.1 V GND
56	F2	—	—	N.C.	—
57	F3	GND	A	VSSHCM	2.9 V GND
58	F4	Power	A	VDDHCM	2.9 V power supply
59	F9	GND	A	VSS33PX	3.3 V GND
60	F10	Power	D	VDDLIF	1.1 V power supply
61	F11	O	D	DOM1	CSI-2 IF output (Data)
62	F12	O	D	DOP1	CSI-2 IF output (Data)
63	G1	Power	D	VDDLSC	1.1 V power supply
64	G2	GND	D	VSSLSC	1.1 V GND
65	G3	GND	A	VSSLCB	1.1 V GND
66	G4	Power	A	VDDLCB	1.1 V power supply
67	G9	—	—	N.C.	—
68	G10	GND	D	VSSLSC	1.1 V GND
69	G11	O	D	DCKM	CSI-2 IF output (Clock)
70	G12	O	D	DCKP	CSI-2 IF output (Clock)
71	H1	I/O	D	SDI / SDA	"4-wire: Serial communication I/F SDI pin I^2C: Serial data line
72	H2	I	D	SLAMODE1	Slave address select
73	H3	—	—	N.C.	—
74	H4	—	—	N.C.	—
75	H9	—	—	N.C.	—
76	H10	Power	D	VDDLSC	1.1 V power supply
77	H11	O	D	DOM0	CSI-2 IF output (Data)
78	H12	O	D	DOP0	CSI-2 IF output (Data)
79	J1	O	D	SDO	4-wire: Serial communication I/F SDO pin I^2C: OPEN
80	J2	I	D	XCLR	System clear (Normal: High, Clear: Low)
81	J3	GND	A	VSS33PX	3.3 V GND
82	J4	Power	A	VDD33PX	3.3 V power supply
83	J5	Power	A	VDDLCB	1.1 V power supply
84	J6	GND	A	VSSHAN	2.9 V GND
85	J7	Power	A	VDD33SUB	3.3 V power supply
86	J8	Power	A	VDD33PX	3.3 V power supply
87	J9	GND	A	VSS33PX	3.3 V GND
88	J10	GND	D	VSSLIF	1.1 V GND
89	J11	O	D	DOM2	CSI-2 IF output (Data)
90	J12	O	D	DOP2	CSI-2 IF output (Data)
91	K1	GND	D	VSSLSC	1.1 V GND
92	K2	I	D	SLAMODE0	Slave address select
93	K3	GND	A	VSSHCM	2.9 V GND
94	K4	Power	A	VDDHCM	2.9 V power supply
95	K5	GND	A	VSSLCB	1.1 V GND
96	K6	—	—	N.C.	—
97	K7	—	—	N.C.	—
98	K8	—	—	N.C.	—
99	K9	—	—	N.C.	—
100	K10	Power	D	VDDLIF	1.1 V power supply
101	K11	—	—	N.C.	—
102	K12	—	—	N.C.	—

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No.	Pin No.	I/O	Analog / Digital	Symbol	Description
103	L1	Power	D	VDDLSC	1.1 V power supply
104	L2	I	D	SLAMODE2	Slave address select
105	L3	I	D	XMASTER	"Master / Slave select (Slave Mode: High, Master Mode: Low)
106	L4	GND	D	VSSLPL	1.1 V GND
107	L5	Power	D	VDDLPL	1.1 V power supply
108	L6	GND	D	VSSLPLA	1.1 V GND
109	L7	Power	D	VDDLPLA	1.1 V power supply
110	L8	GND	D	VSSLPLA	1.1 V GND
111	L9	Power	D	VDDLPLA	1.1 V power supply
112	L10	GND	D	VSSLIF	1.1 V GND
113	L11	—	—	N.C.	—
114	L12	—	—	N.C.	—
115	M3	Power	D	VDDLCN	1.1 V power supply
116	M4	GND	D	VSSLCN	1.1 V GND
117	M5	O	D	TOUT0	Pulse0 output pin
118	M6	O	D	TOUT2	Pulse2 output pin
119	M7	O	D	TOUT1	Pulse1 output pin
120	M8	GND	D	VSSLPLD	1.1 V GND
121	M9	Power	D	VDDLPLD	1.1 V power supply
122	M10	GND	D	VSSLSC	1.1 V GND
123	N1	—	—	N.C.	GND connection is allowed
124	N3	Power	D	VDDLSC	1.1 V power supply
125	N4	GND	D	VSSLSC	1.1 V GND
126	N5	I/O	D	XHS	Horizontal sync signal
127	N6	I	D	SCK / SCL	4-wire: Serial communication I/F SCK pin I^2C: Serial clock line
128	N7	I	D	XCE	4-wire: Serial communication I/F XCE pin I^2C: Fixed to High
129	N8	I	D	INCK	Master clock input
130	N9	Power	D	VDDMIO	1.8 V power supply
131	N10	Power	D	VDDLSC	1.1 V power supply
132	N12	—	—	N.C.	GND connection is allowed

* N.C. pins in the table above should be left open on the board.

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11. Electrical Characteristics

11.1. DC Characteristics

Item		Pins	Symbol	Conditions	Min.	Typ.	Max.	Unit
Supply voltage	Analog	V_{DD}33x	AV_{DD}1	—	3.15	3.3	3.45	V
	Analog	V_{DD}Hx	AV_{DD}2	—	2.75	2.9	3.05	V
	Digital	V_{DD}Mx	OV_{DD}	—	1.7	1.8	1.9	V
	Digital	V_{DD}Lx	DV_{DD}	—	1.0	1.1	1.2	V
Digital input voltage		XHS XVS XCLR INCK XMASTER SLAMODE0 SLAMODE1 SLAMODE2 SCK SDI XCE XTRIG1 XTRIG2	VIH	XVS / XHS in Slave mode	0.7 × OV_{DD}	—	—	V
			VIL		—	—	0.3 × OV_{DD}	V
Digital output voltage		XHS XVS SDO TOUT0 TOUT1 TOUT2	VOH	XVS / XHS in Master mode	OV_{DD} – 0.2	—	—	V
			VOL		—	—	0.2	V

11.2. Power Consumption

Item	Pins	Symbol	Typ.	Max.	Unit
Operating current MIPI CSI-2 4 Lane 12 bit 67.2 frame/s	V_{DD}33	IAV_{DD}1	28	45	mA
	V_{DD}H	IAV_{DD}2	79	125	mA
	V_{DD}M	IOV_{DD}	0.1	2	mA
	V_{DD}L	IDV_{DD}	153	265	mA
Standby current	V_{DD}33	IAV_{DD}1	—	0.1	mA
	V_{DD}H	IAV_{DD}2	—	0.1	mA
	V_{DD}M	IOV_{DD}_STB	—	0.05	mA
	V_{DD}L	IDV_{DD}_STB	—	200	mA

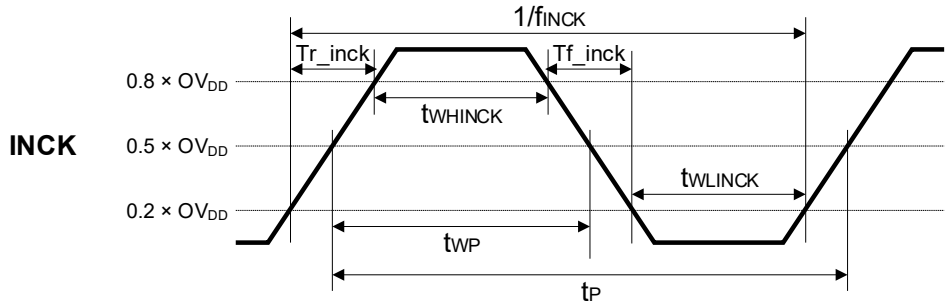
Operating current:
(Typical value condition) : Supply voltage: 3.3 V / 2.9 V / 1.8 V / 1.1 V, Tj = 25 ˚C
(Maximum value condition) : Supply voltage: 3.45 V / 3.05 V / 1.9 V / 1.2 V, Tj = 60 ˚C
Worst state of internal circuit operating current consumption.

Standby current:
(Maximum value condition) : Supply voltage: 3.45 V / 3.05 V / 1.9 V / 1.2 V, Tj = 60 ˚C, INCK = 0 V,
The device in the light-obstructed state.

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11.3. AC Characteristics

11.3.1. Master Clock (INCK)



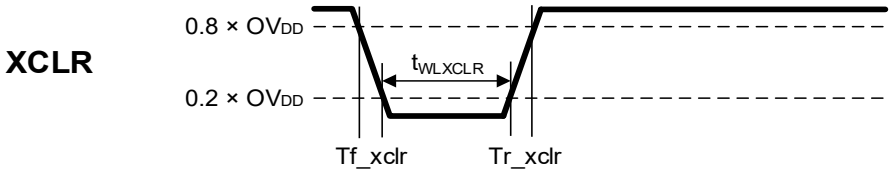
Duty Ratio = t_{WP} / t_P × 100

Item	Symbol	Min.	Typ.	Max.	Unit	Remarks
INCK clock frequency	f_{INCK}	f_{INCK} × 0.96	f_{INCK}	f_{INCK} × 1.02	MHz	f_{INCK} = 37.125 MHz, 74.25 MHz, 54 MHz
INCK Low level pulse width	t_{WLINCK}	4	—	—	ns	
INCK High level pulse width	t_{WHINCK}	4	—	—	ns	
INCK clock duty	—	40.0	50.0	60.0	%	Define with 0.5 × OV_{DD}
INCK Rise time	Tr_inck	—	—	5	ns	20 % to 80 %
INCK Fall time	Tf_inck	—	—	5	ns	20 % to 80 %

* The INCK fluctuation affects the frame rate.

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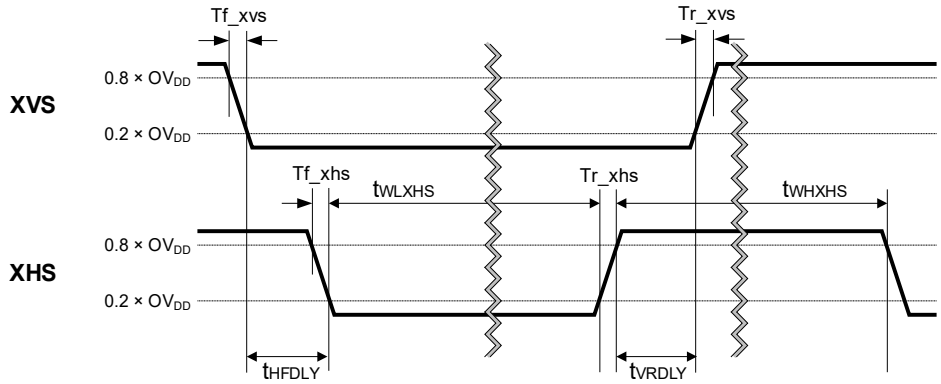
11.3.2. System Clear (XCLR)



Item	Symbol	Min.	Typ.	Max.	Unit	Remarks
XCLR Low level pulse width	t_{WLXCLR}	$4/f_{INCK}$	—	—	ns	f_{INCK} = 37.125 MHz, 74.25 MHz, 54 MHz
XCLR Fall time	Tf_xclr	—	—	5	ns	80 % to 20 %
XCLR Rise time	Tr_xclr	—	—	5	ns	20 % to 80 %

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11.3.3. XVS / XHS Input Characteristics in Slave Mode (XMASTER = High)

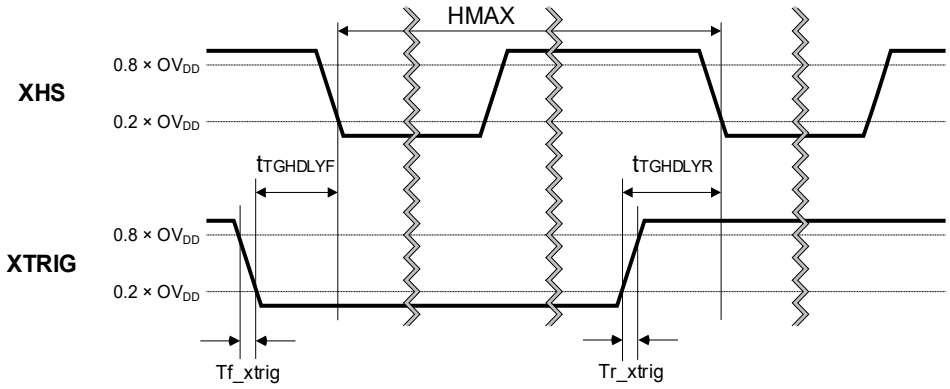


Item	Symbol	Min.	Typ.	Max.	Unit	Remarks
XHS Low level pulse width	t_{WLXHS}	$4/f_{INCK}$	—	—	ns	
XHS High level pulse width	t_{WHXHS}	$4/f_{INCK}$	—	—	ns	
XVS - XHS fall width	t_{HFDLY}	$1/f_{INCK}$	—	—	ns	
XHS - XVS rise width	t_{VRDLY}	$1/f_{INCK}$	—	—	ns	
XVS Fall time	Tf_xvs	—	—	5	ns	80 % to 20 %
XVS Rise time	Tr_xvs	—	—	5	ns	20 % to 80 %
XHS Fall time	Tf_xhs	—	—	5	ns	80 % to 20 %
XHS Rise time	Tr_xhs	—	—	5	ns	20 % to 80 %

Synchronization cannot be performed from XVS and XHS signal in mater mode. Detect the packet header output from the image sensor.

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11.3.4. XTRIG Input Characteristics in Slave Mode (XMASTER = High) only

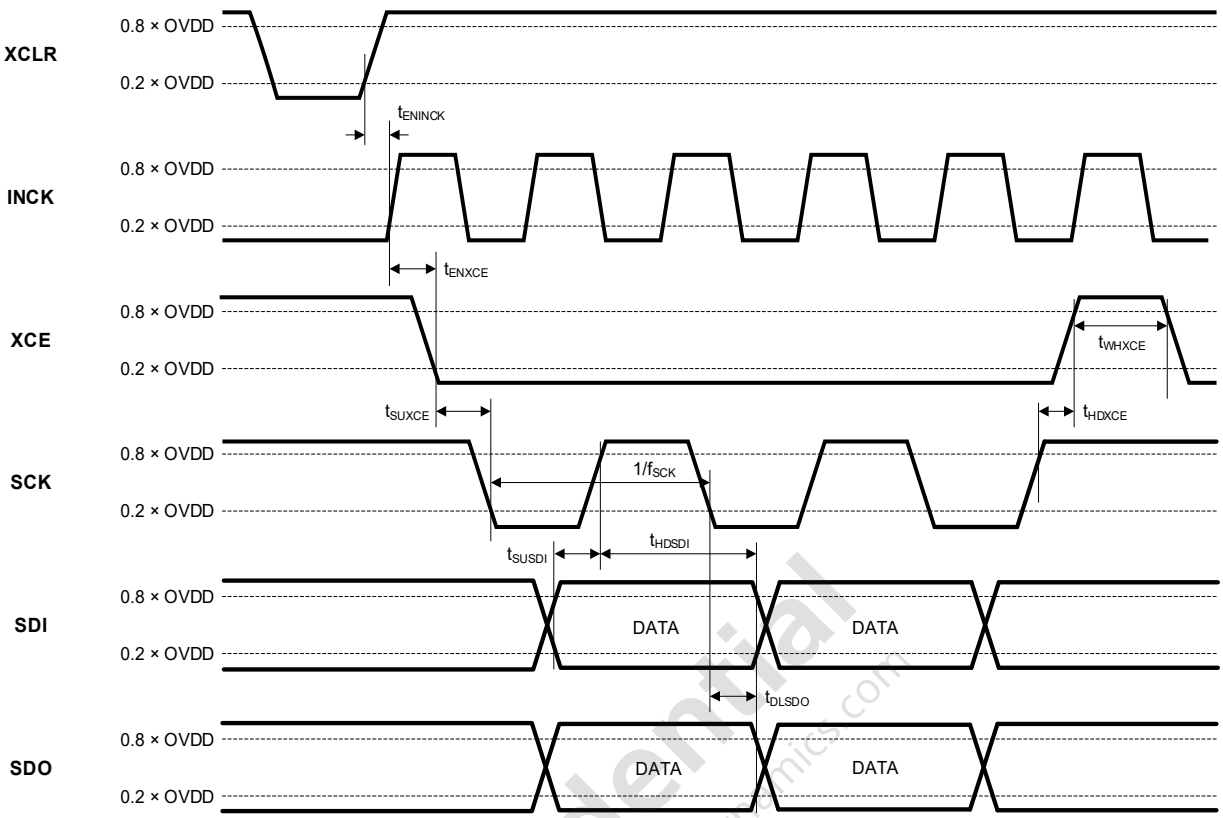


Item	Symbol	Min.	Typ.	Max.	Unit	Remarks
XTRIG fall - XHS fall width	$t_{TGHDLYF}$	10	—	HMAX-10	INCK	
XTRIG rise - XHS fall width	$t_{TGHDLYR}$	10	—	HMAX-10	INCK	
XTRIG Fall time	Tf_xtrig	—	—	5	ns	80 % to 20 %
XTRIG Rise time	Tr_xtrig	—	—	5	ns	20 % to 80 %

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11.3.5. Serial Communication

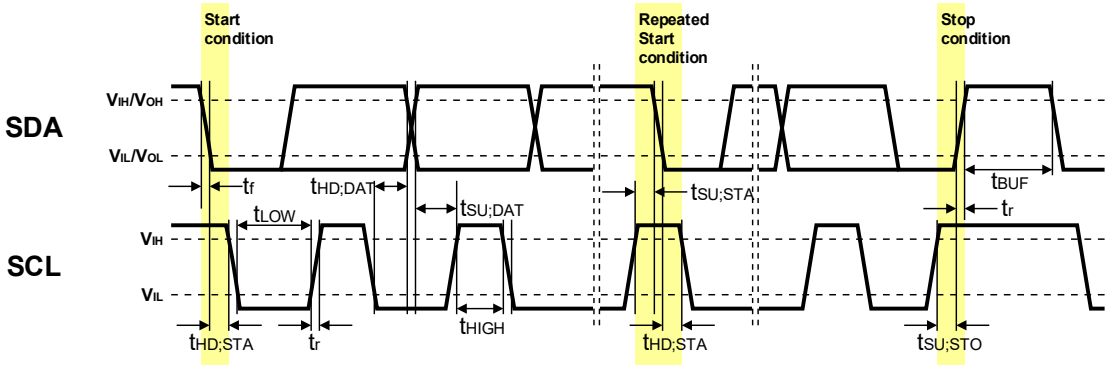
4-wire



Item	Symbol	Min.	Typ.	Max.	Unit	Remarks
SCK clock frequency	f_{SCK}	—	—	13.5	MHz	
INCK effective margin	t_{ENINCK}	1	—	—	μs	
XCE effective margin	t_{ENXCE}	20	—	—	μs	
XCE input setup time	t_{SUXCE}	20	—	—	ns	
XCE input hold time	t_{HDXCE}	20	—	—	ns	
XCE High level pulse width	t_{WHXCE}	20	—	—	ns	
SDI input setup time	t_{SUSDI}	10	—	—	ns	
SDI input hold time	t_{HDSDI}	10	—	—	ns	
SDO output delay time	t_{DLSDO}	0	—	25	ns	Output load capacitance: 20 pF

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I²C



I²C Specification

Item	Symbol	Min.	Typ.	Max.	Unit	Remarks
Low level input voltage	V_{IL}	-0.3	—	0.3 × OV_{DD}	V	
High level input voltage	V_{IH}	0.7 × OV_{DD}	—	1.9	V	
Low level output voltage	V_{OL}	0	—	0.2 × OV_{DD}	V	OV_{DD} < 2 V, Sink 3 mA
High level output voltage	V_{OH}	0.8 × OV_{DD}	—	—	V	
Output fall time from VI_{Hmin} to V_{ILmax}	tof	—	—	250	ns	Load 10 pF – 400 pF, 0.7 × OV_{DD} – 0.3 × OV_{DD}
Input current	Ii	-10	—	10	µA	0.1 × OV_{DD} – 0.9 × OV_{DD}
Capacitance for SCK (/SCL) , SDI (/SDA)	Ci	—	—	10	pF	

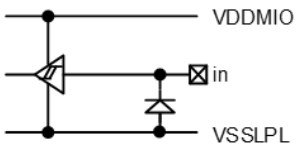
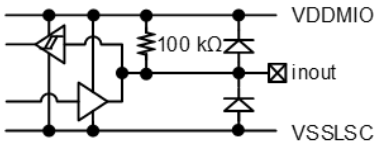
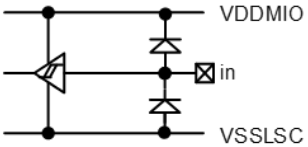
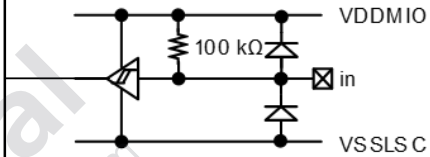
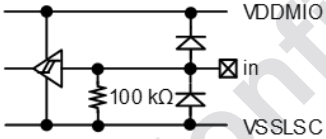
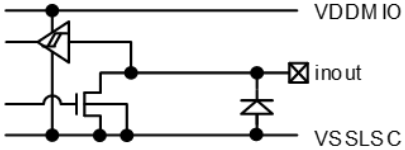
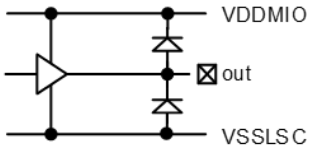
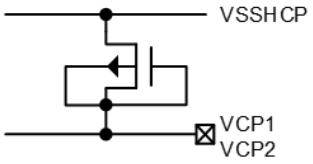
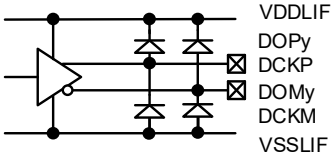
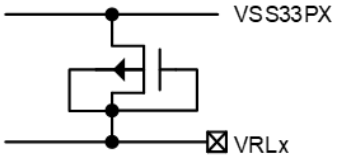
I²C AC Characteristics

Item	Symbol	Min.	Typ.	Max.	Unit
SCL clock frequency	f_{SCL}	0	—	400	kHz
Hold time (Start Condition)	t_{HDSTA}	0.6	—	—	µs
Low period of the SCL clock	t_{LOW}	1.3	—	—	µs
High period of the SCL clock	t_{HIGH}	0.6	—	—	µs
Set-up time (Repeated Start Condition)	t_{SUSTA}	0.6	—	—	µs
Data hold time	t_{HDDAT}	0	—	0.9	µs
Data set-up time	t_{SUDAT}	100	—	—	ns
Rise time of both SDA and SCL signals	t_R	—	—	300	ns
Fall time of both SDA and SCL signals	t_F	—	—	300	ns
Set-up time (Stop Condition)	t_{SUSTO}	0.6	—	—	µs
Bus free time between a Stop and Start Condition	t_{BUF}	1.3	—	—	µs

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12. I/O Equivalent Circuit Diagram

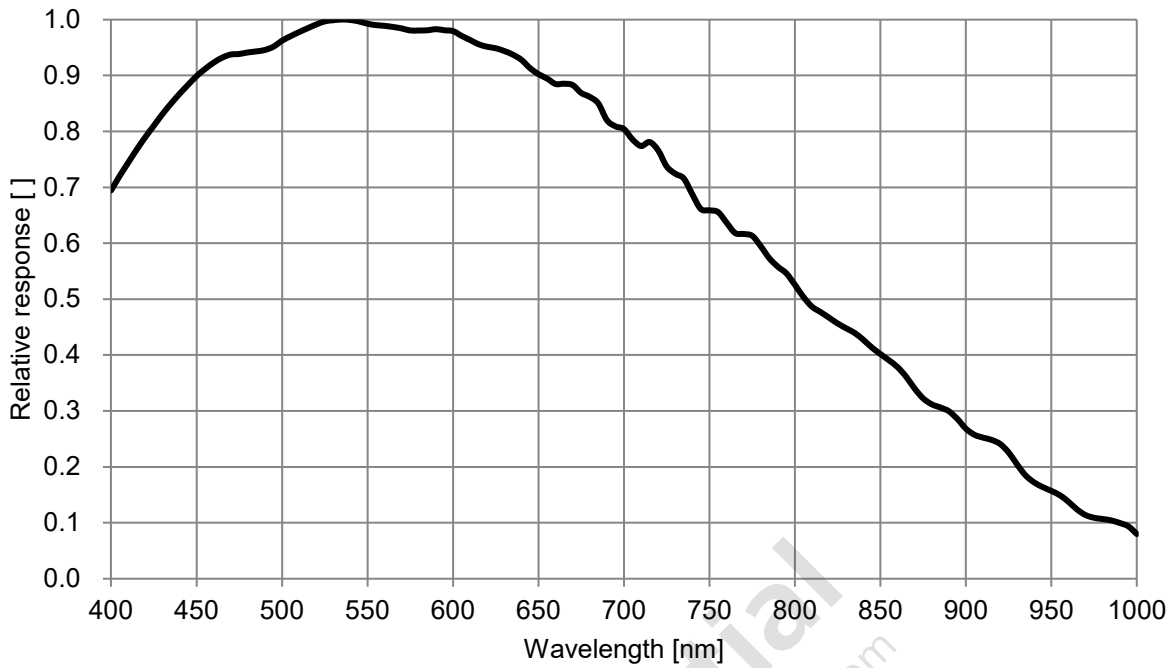
□ : External pin

Symbol	Equivalent circuit	Symbol	Equivalent circuit
INCK		XVS XHS	
XCLR XCE XMASTER		XTRIG1, 2	
SLAMODE 0, 1, 2		SDI / SDA SCK / SCL	
SDO		VCP1 VCP2	
DOPy DOMy DCKP DCKM y : 0 to 3		VRLFDB VRLOFG VRLSEL VRLTRG VRLTRX VRLTRY	 VRLx: VRLFDB, VRLOFG, VRLSEL, VRLTRG, VRLTRX and VRLTRY

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13. Spectral Sensitivity Characteristics

(Characteristics in the wafer status)



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14. Setting Registers Using Serial Communication

This sensor can write and read the setting values of the various registers shown in the Register Map by 4-wire serial communication and I²C communication. See the Register Map for the addresses and setting values to be set. Because the two communication systems are judged at the first communication, once they are judged, the communication cannot be switched until sensor reset. The pin for 4-wire serial communication and I²C communication is shared, so the external pin XCE must be fixed to power supply side when using I²C communication.

14.1. Description of Setting Registers (4-wire)

The serial data input order is LSB-first transfer. The table below shows the various data types and descriptions.

Serial Data Transfer Order

Chip ID	Start address	Data	Data	Data	…
(8 bit)	(8 bit)	(8 bit)	(8 bit)	(8 bit)	(8 bit)

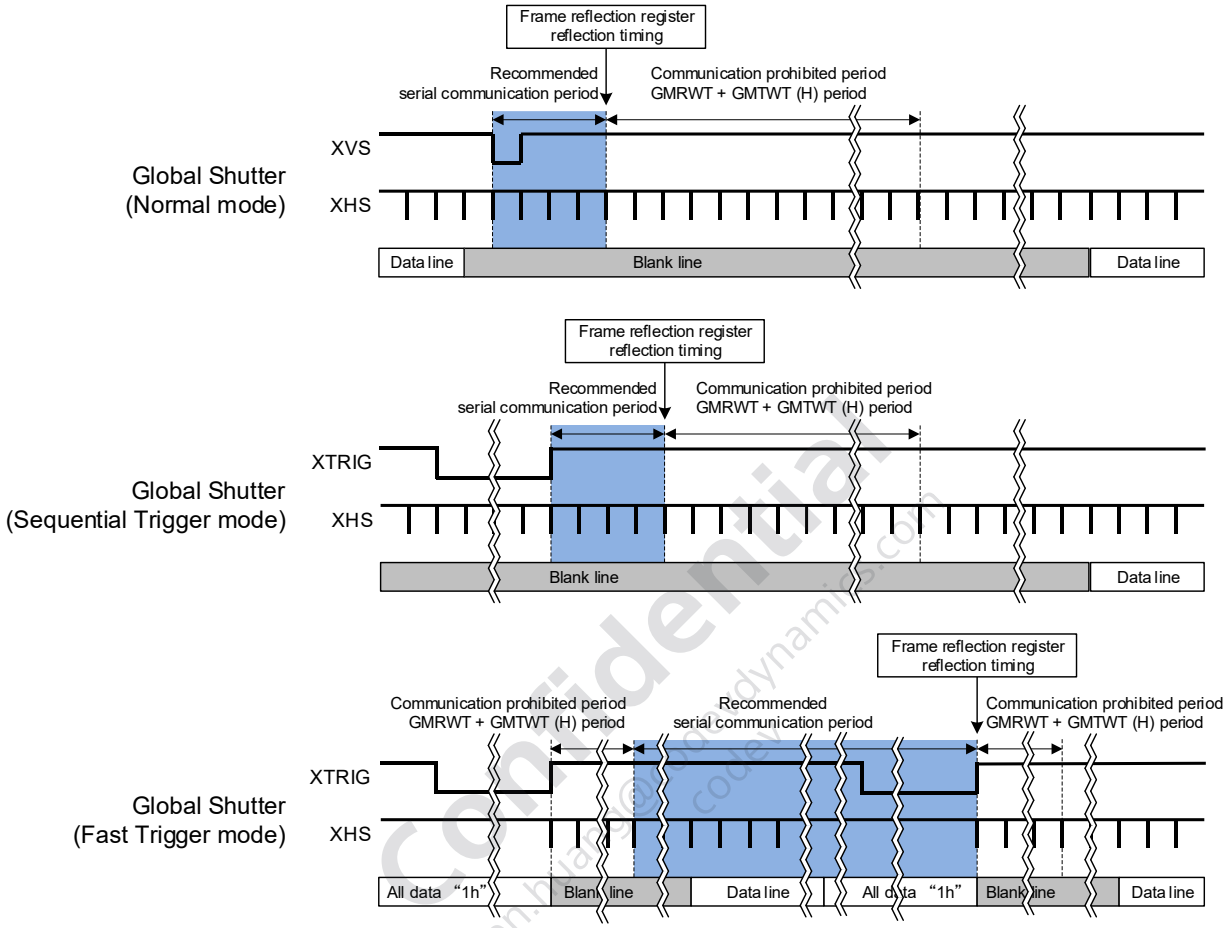
Type and Description

Type	Description
Chip ID	Chip ID: 02 Write: 02h / Read: 82h
	Chip ID: 03 Write: 03h / Read: 83h
	Chip ID: 04 Write: 04h / Read: 84h
	Chip ID: 05 Write: 05h / Read: 85h
	Chip ID: 06 Write: 06h / Read: 86h
	Chip ID: 07 Write: 07h / Read: 87h
	Chip ID: 08 Write: 08h / Read: 88h
	Chip ID: 09 Write: 09h / Read: 89h
	Chip ID: 0A Write: 0Ah / Read: 8Ah
	Chip ID: 0B Write: 0Bh / Read: 8Bh
	Chip ID: 0C Write: 0Ch / Read: 8Ch
	Chip ID: 0D Write: 0Dh / Read: 8Dh
	Chip ID: 0E Write: 0Eh / Read: 8Eh
	Chip ID: 10 Write: 10h / Read: 90h
	Chip ID: 11 Write: 11h / Read: 91h
	Chip ID: 12 Write: 12h / Read: 92h
	Chip ID: 13 Write: 13h / Read: 93h
	Chip ID: 14 Write: 14h / Read: 94h
	Chip ID: 15 Write: 15h / Read: 95h
	Chip ID: 16 Write: 16h / Read: 96h
	Chip ID: 17 Write: 17h / Read: 97h
	Chip ID: 18 Write: 18h / Read: 98h
	Chip ID: 19 Write: 19h / Read: 99h
	Chip ID: 1A Write: 1Ah / Read: 9Ah
	Chip ID: 1B Write: 1Bh / Read: 9Bh
	Chip ID: 1C Write: 1Ch / Read: 9Ch
	Chip ID: 1E Write: 1Eh / Read: 9Eh
Address	Designate the address according to the Register Map. When using a communication method that designates continuous addresses, the address is automatically incremented from the previously transmitted address.
Data	Input the setting values according to the Register Map.

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14.2. Register Communication Timing (4-wire)

Perform serial communication in sensor standby mode or within communication period. For the registers marked "V" in the item of Reflection timing, when the communication is performed in the communication period shown in the figure below they are reflected by frame reflection timing. For the registers noted “Immediately” in the item of Reflection timing, the settings are reflected when the communication is performed.
For GMRWT and GMTWT, refer to the register list of each readout mode.



Communication Timing and Register Reflection Timing

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14.3. Register Write and Read (4-wire)

◆ Follow the communication procedure below when writing registers.

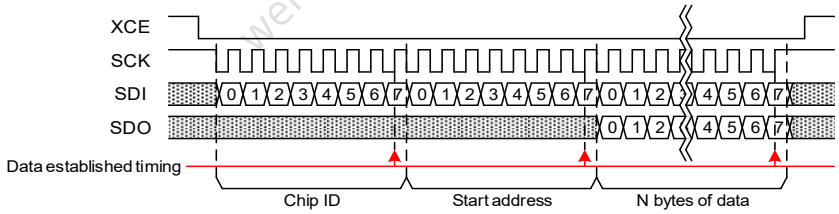
(1) Set XCE Low to enable the chip's communication function. Serial data input is executed using SCK and SDI.
(2) Transmit data in sync with SCK 1 bit at a time from the LSB using SDI. Transfer SDI in sync with the falling edge of SCK. (The data is loaded at the rising edge of SCK.)
(3) Input the Chip ID (CID = 02h to 0Eh, 10h to 1Ch, 1Eh) to the first byte. If the Chip ID differs, subsequent data is ignored.
(4) Input the start address to the second byte. The address is automatically incremented.
(5) Input the data to the third and subsequent bytes. The data in the third byte is written to the register address designated by the second byte, and the register address is automatically incremented thereafter when writing the data for the fourth and subsequent bytes. Normal register data is loaded to the inside of the sensor and established in 8-bit units.
(6) The register values starting from the register address designated by the second byte are output from the SDO pin. The register values before the write operation are output. The actual register values are the input data.
(7) Set XCE High to end communication.

◆ Follow the communication procedure below when reading registers.

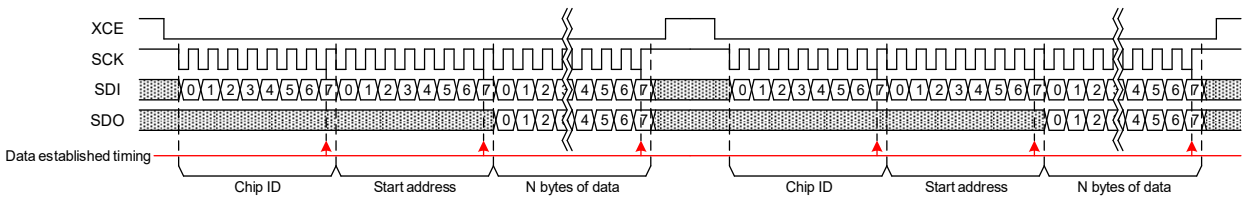
(1) Set XCE Low to enable the chip's communication function. Serial data input is executed using SCK and SDI.
(2) Transmit data in sync with SCK 1 bit at a time from the LSB using SDI. Transfer SDI in sync with the falling edge of SCK. (The data is loaded at the rising edge of SCK.)
(3) Input Chip ID (CID = 82h to 8Eh, 90h to 9Ch, 9Eh) to the first byte. If the Chip ID differs, subsequent data is ignored.
(4) Input the start address to the second byte. The address is automatically incremented.
(5) Input data to the third and subsequent bytes. Input dummy data in order to read the registers. The dummy data is not written to the registers. To read continuous data, input the necessary number of bytes of dummy data.
(6) The register values starting from the register address designated by the second byte are output from the SDO pin. The input data is not written, so the actual register values are output.
(7) Set XCE High to end communication.

Note) When writing data to multiple registers with discontinuous addresses, access to undesired registers can be avoided by repeating the above procedure multiple times.

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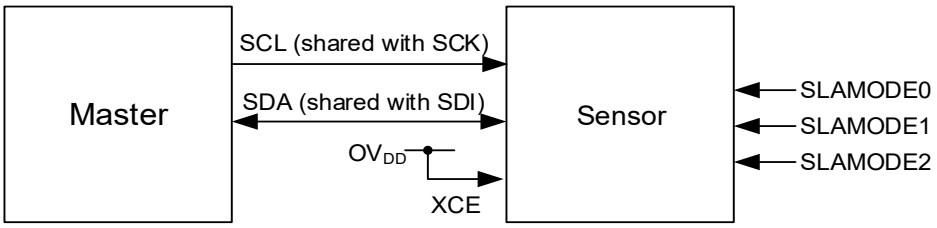
Serial Communication (Continuous Addresses)



Serial Communication (Discontinuous Addresses)

14.4. Description of Setting Registers (I^2C)

The serial data input order is MSB-first transfer. The table below shows the various data types and descriptions.



Pin connection of serial communication

The sensor can use two kinds of slave addresses by switching the polarity of SLAMODE0, SLAMODE1 and SLAMODE2 Pins for one I^2C bus, and can use a common slave address in both polarities of SLAMODE [2:0] Pins for one I^2C bus.

1st SLAVE Address (SLAMODE [2:0])

SLAMODE [2:0]	MSB							LSB
0h	0	1	1	0	1	1	0	R / W
1h	0	1	1	0	1	1	1	R / W
2h	0	1	1	1	0	0	0	R / W
3h	0	1	1	1	0	0	1	R / W
4h	0	1	1	1	0	1	0	R / W
5h	0	1	1	1	0	1	1	R / W
6h	0	1	1	1	1	0	0	R / W
7h	0	1	1	1	1	0	1	R / W

2nd SLAVE Address (Common in all SLAMODE [2:0])

MSB							LSB
0	0	1	1	0	1	0	R / W

* R/W is data direction bit

R/W

R / W bit	Data direction
0	Write (Master → Sensor)
1	Read (Sensor → Master)

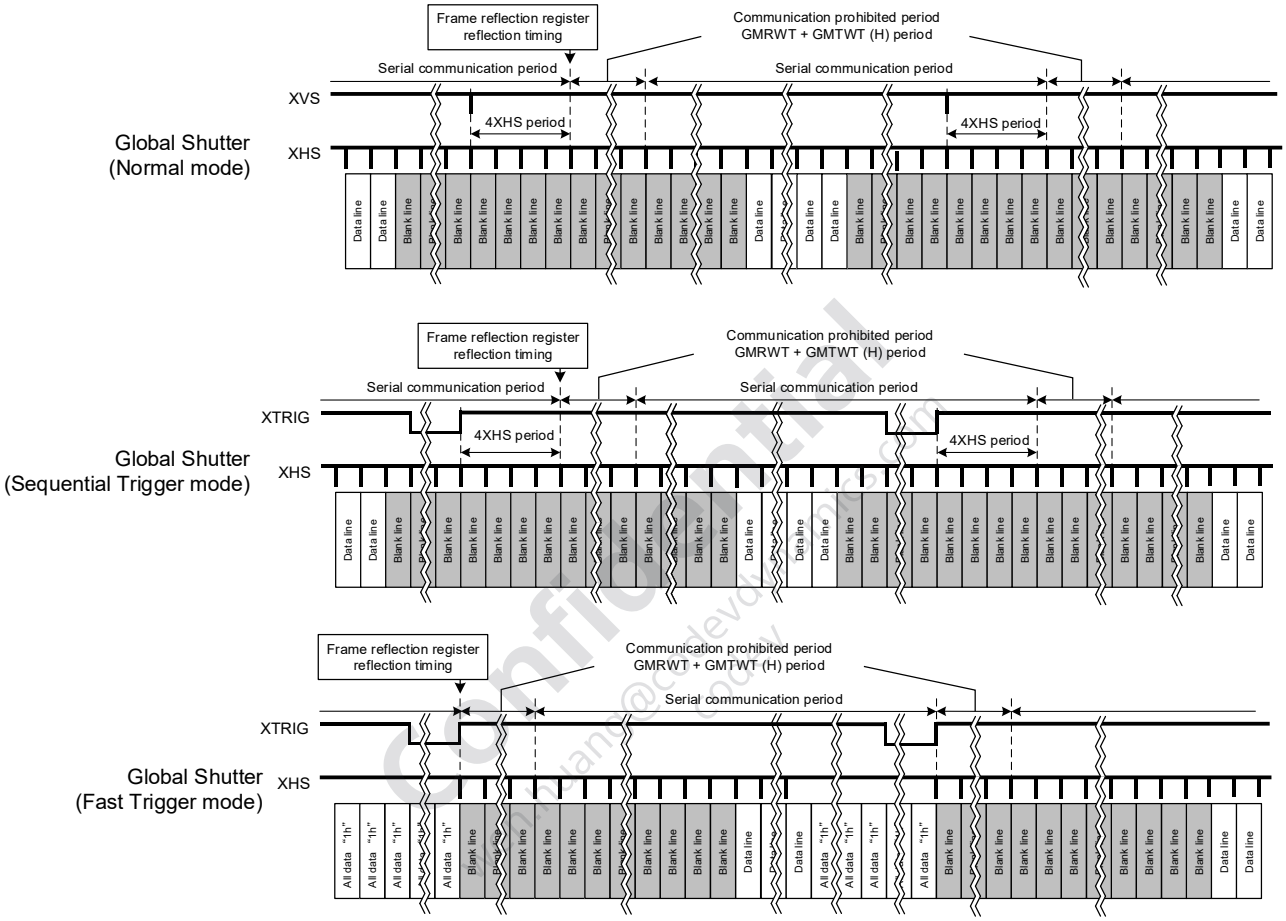
I^2C pin description

Symbol	Pin No.	Description
SCL (common to SCK)	N6	Serial clock input
SDA (common to SDI)	H1	Serial data communication
SLAMODE0	K2	I^2C 1st Slave address select
SLAMODE1	H2	I^2C 1st Slave address select
SLAMODE2	L2	I^2C 1st Slave address select
XCE	N7	Fixed to High when using I^2C

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14.5. Register Communication Timing (I²C)

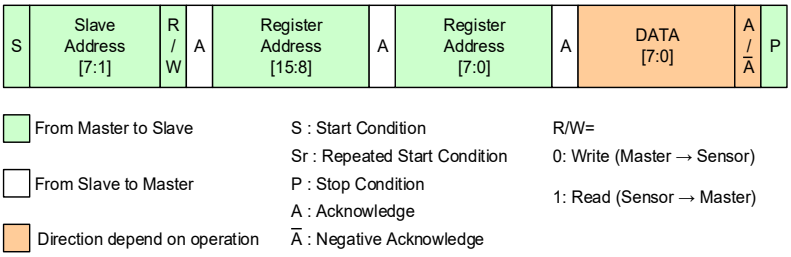
In I²C communication system, communication can be performed excluding the communication prohibition period (GMRWT + GMTWT (H) period) shown below.
For the registers marked "V" in the item of Reflection timing, when the communication is performed in the communication period shown in the figure below they are reflected by frame reflection timing. For the registers noted “Immediately” in the item of Reflection timing, the settings are reflected when the communication is performed. Using REGHOLD function is recommended for register setting using I²C communication. For REGHOLD function, see “Register Hold Setting” in chapter “Description of Various Functions”.
For GMRWT and GMTWT, refer to the register list of each readout mode.



Communication Timing and Register Reflection Timing

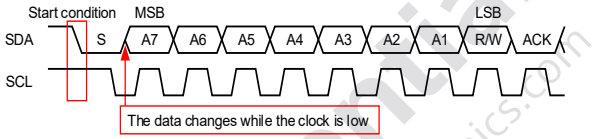
14.6. I²C Communication Protocol

I²C serial communication supports a 16-bit register address and 8-bit data message type.

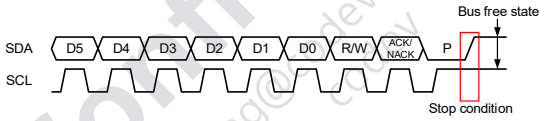


Communication protocol

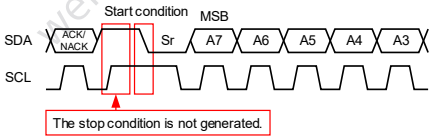
Data is transferred serially, MSB first in 8-bit units. After each data byte is transferred, A (Acknowledge) / Ā (Negative Acknowledge) is transferred. Data (SDA) is transferred at the clock (SCL) cycle. SDA can change only while SCL is Low, so the SDA value must be held while SCL is High. The Start Condition is defined by SDA changing from High to Low while SCL is High. When the Stop Condition is not generated in the previous communication phase and Start Condition for the next communication is generated, that Start Condition is recognized as a Repeated Start Condition.



Start Condition

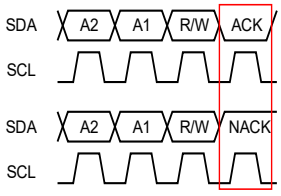


Stop Condition



Repeated Start Condition

After transfer of each data byte, the Master or the sensor transmits an Acknowledge / Negative Acknowledge and release (does not drive) SDA. When Negative Acknowledge is generated, the Master must immediately generate the Stop Condition and end the communication.



Acknowledge and Negative Acknowledge

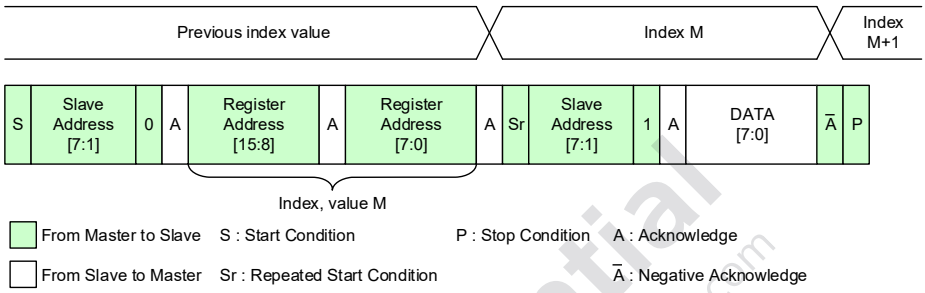
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14.7. I²C Serial Communication Read/Write Operation

This sensor supports the following four read operations and two write operations.

14.7.1. Single Read from Random Location

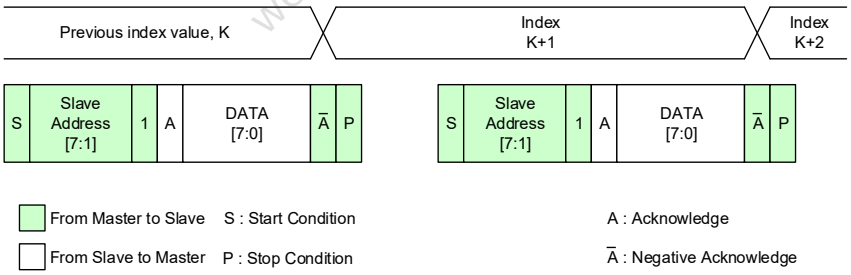
The sensor has an index function that indicates which address it is focusing on. In reading the data at an optional single address, the Master must set the index value to the address to be read. For this purpose it performs dummy write operation up to the register address. The upper level of the figure below shows the sensor internal index value, and the lower level of the figure shows the SDA I/O data flow. The Master sets the sensor index value to M by designating the sensor slave address with a write request, then designating the address (M). Then, the Master generates the Start Condition. The Start Condition is generated without generating the Stop Condition, so it becomes the Repeated Start Condition. Next, when the Master sends the slave address with a read request, the sensor outputs an Acknowledge immediately followed by the index address data on SDA. After the Master receives the data, it generates a Negative Acknowledge and the Stop Condition to end the communication.



Single Read from Random Location

14.7.2. Single Read from Current Location

After the slave address is transmitted by a write request, that address is designated by the next communication and the index holds that value. In addition, when data read/write is performed, the index is incremented by the subsequent Acknowledge/Negative Acknowledge timing. When the index value is known to indicate the address to be read, sending the slave address with a read request allows the data to be read immediately after Acknowledge. After receiving the data, the Master generates a Negative Acknowledge and the Stop Condition to end the communication, but the index value is incremented, so the data at the next address can be read by sending the slave address with a read request.

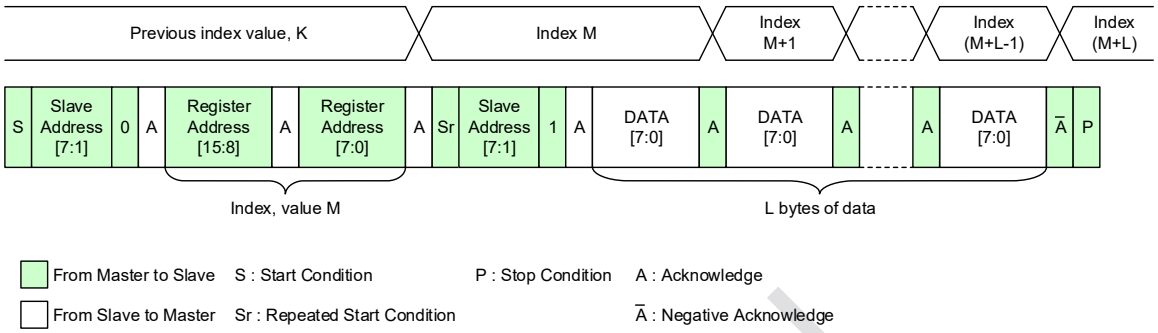


Single Read from Current Location

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14.7.3. Sequential Read Starting from Random Location

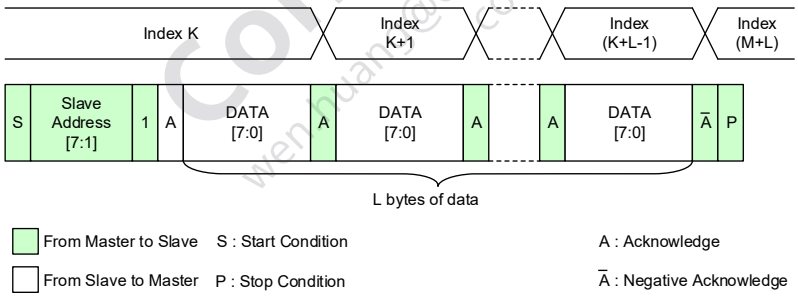
In reading data sequentially, which is starting from an optional address, the Master must set the index value to the start of the addresses to be read. For this purpose, dummy write operation includes the register address setting. The Master sets the sensor index value to M by designating the sensor slave address with a read request, then designating the address (M). Then, the Master generates the Repeated Start Condition. Next, when the Master sends the slave address with a read request, the sensor outputs an Acknowledge followed immediately by the index address data on SDA. When the Master outputs an Acknowledge after it receives the data, the index value inside the sensor is incremented and the data at the next address is output on SDA. This allows the Master to read data sequentially. After reading the necessary data, the Master generates a Negative Acknowledge and the Stop Condition to end the communication.



Sequential Read Starting from Random Location

14.7.4. Sequential Read Starting from Current Location

When the index value is known to indicate the address to be read, sending the slave address with a read request allows the data to be read immediately after the Acknowledge. When the Master outputs an Acknowledge after it receives the data, the index value inside the sensor is incremented and the data at the next address is output on SDA. This allows the Master to read data sequentially. After reading the necessary data, the Master generates a Negative Acknowledge and the Stop Condition to end the communication.

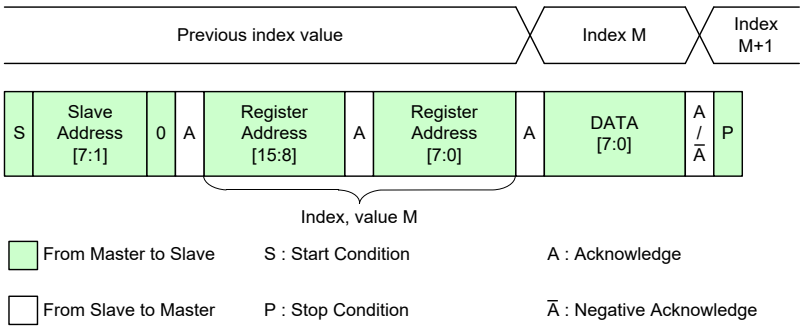


Sequential Read Starting from Current Location

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14.7.5. Single Write to Random Location

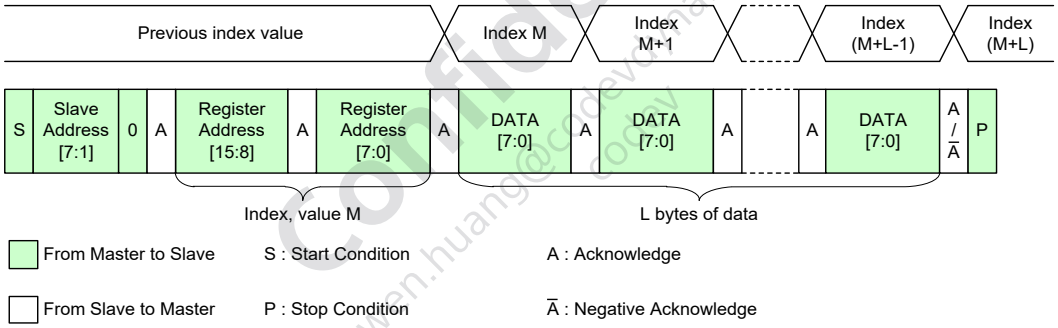
The Master sets the sensor index value to M by designating the sensor slave address with a write request, and designating the address (M). After that the Master can write the value in the designated register by transmitting the data to be written. After writing the necessary data, the Master generates the Stop Condition to end the communication.



Single Write to Random Location

14.7.6. Sequential Write Starting from Random Location

The Master can write a value to register address M by designating the sensor slave address with a write request, designating the address (M), and then transmitting the data to be written. After the sensor receives the write data, it outputs an Acknowledge and at the same time increments the register address, so the Master can write to the next address simply by continuing to transmit data. After the Master writes the necessary number of bytes, it generates the Stop Condition to end the communication.



Sequential Write Starting from Random Location

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15. Readout Drive Modes

The table below lists the operating modes available with this sensor. (Each value is the Max. frame rate of each number of Lane.)

Data rate = 1188 Mbps

Drive mode	CSI-2 Lane[*1]	Data rate [Mbps/Lane]	A/D conversion	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	1188	8	96.2	2448	2048	174.0	348.0	253.1
	2	1188		51.7			328.0	656.0	477.1
	4	1188	10	79.1			212.5	425.0	309.1
	2	1188		42.0			405.0	810.0	589.1
	4	1188	12	67.2			251.0	502.0	365.1
	2	1188		35.3			482.5	965.0	701.9
Vertical / Horizontal 1/2 subsampling	4	1188	8	316.8	1224	1024	97.0	194.0	141.1
	2	1188		183.9			174.0	348.0	253.1
	4	1188	10	268.4			116.0	232.0	168.8
	2	1188		152.1			212.5	425.0	309.1
	4	1188	12	232.9			135.5	271.0	197.1
	2	1188		129.4			251.5	503.0	365.9
2 × 2 FD binning mode	4	1188	8	316.8	1224	1024	97.0	194.0	141.1
	2	1188		183.9			174.0	348.0	253.1
	4	1188	10	268.4			116.0	232.0	168.8
	2	1188		152.1			212.5	425.0	309.1
	4	1188	12	232.9			135.5	271.0	197.1
	2	1188		129.4			251.5	503.0	365.9
ROI	4	1188	8	[*2]	[*3]	[*3]	174.0	348.0	253.1
	2	1188		[*2]			328.0	656.0	477.1
	4	1188	10	[*2]			212.5	425.0	309.1
	2	1188		[*2]			405.0	810.0	589.1
	4	1188	12	[*2]			251.0	502.0	365.1
	2	1188		[*2]			482.5	965.0	701.9

[*1] The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 96.2 [frame/s] mode: 1188 [Mbps/Lane] × 4 = 4.752 [Gbps]

[*2] See the section of “ROI mode”

[*3] Designated cropping area (ROI)

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Data rate = 891 Mbps

Drive mode	CSI-2 Lane[*1]	Data rate [Mbps/Lane]	A/D conversion	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	891	8	73.3	2448	2048	230.0	460.0	334.6
	2	891		39.1			435.0	870.0	632.8
	4	891	10	60.2			281.0	562.0	408.8
	2	891		31.8			537.0	1074.0	781.1
	4	891	12	51.1			332.0	664.0	483.0
	2	891		26.7			640.0	1280.0	931.0
Vertical / Horizontal 1/2 subsampling	4	891	8	248.5	1224	1024	127.0	254.0	184.8
	2	891		141.4			229.5	459.0	333.9
	4	891	10	209.1			152.5	305.0	221.9
	2	891		117.1			280.0	560.0	407.3
	4	891	12	180.4			178.0	356.0	259.0
	2	891		99.1			332.0	664.0	483.0
2 × 2 FD binning mode	4	891	8	248.5	1224	1024	127.0	254.0	184.8
	2	891		141.4			229.5	459.0	333.9
	4	891	10	209.1			152.5	305.0	221.9
	2	891		117.1			280.0	560.0	407.3
	4	891	12	180.4			178.0	356.0	259.0
	2	891		99.1			332.0	664.0	483.0
ROI	4	891	8	*2	*3	*3	230.0	460.0	334.6
	2	891		*2			435.0	870.0	632.8
	4	891	10	*2			281.0	562.0	408.8
	2	891		*2			537.0	1074.0	781.1
	4	891	12	*2			332.0	664.0	483.0
	2	891		*2			640.0	1280.0	931.0

*1 The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 73.3 [frame/s] mode: 891 [Mbps/Lane] × 4 = 3.564 [Gbps]

*2 See the section of “ROI mode”

*3 Designated cropping area (ROI)

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Data rate = 594 Mbps

Drive mode	CSI-2 Lane[*1]	Data rate [Mbps/Lane]	A/D conversion	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	594	8	49.8	2448	2048	341.0	682.0	496.0
	2	594		26.3			648.5	1297.0	943.3
	4	594	10	40.7			417.5	835.0	607.3
	2	594		21.3			801.5	1603.0	1165.9
	4	594	12	34.5			494.5	989.0	719.3
	2	594		17.9			956.0	1912.0	1390.6
Vertical / Horizontal 1/2 subsampling	4	594	8	171.7	1224	1024	187.0	374.0	272.0
	2	594		96.6			340.5	681.0	495.3
	4	594	10	144.2			225.0	450.0	327.3
	2	594		79.5			416.5	833.0	605.9
	4	594	12	124.4			263.5	527.0	383.3
	2	594		67.3			494.0	988.0	718.6
2 × 2 FD binning mode	4	594	8	171.7	1224	1024	187.0	374.0	272.0
	2	594		96.6			340.5	681.0	495.3
	4	594	10	144.2			225.0	450.0	327.3
	2	594		79.5			416.5	833.0	605.9
	4	594	12	124.4			263.5	527.0	383.3
	2	594		67.3			494.0	988.0	718.6
ROI	4	594	8	*2	*3	*3	341.0	682.0	496.0
	2	594		*2			648.5	1297.0	943.3
	4	594	10	*2			417.5	835.0	607.3
	2	594		*2			801.5	1603.0	1165.9
	4	594	12	*2			494.5	989.0	719.3
	2	594		*2			956.0	1912.0	1390.6

[*1] The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 49.8 [frame/s] mode: 594 [Mbps/Lane] × 4 = 2.376 [Gbps]

[*2] See the section of “ROI mode”

[*3] Designated cropping area (ROI)

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16. Description of Various Functions

16.1. Standby mode

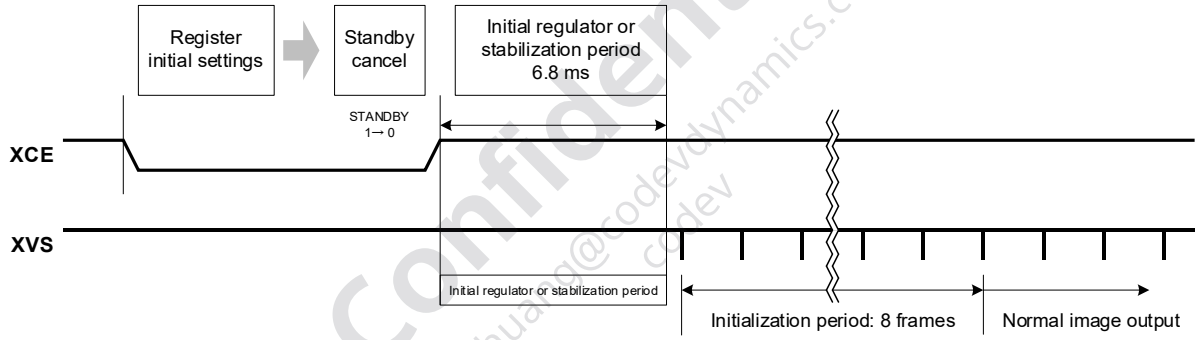
This sensor stops its operation and goes into standby mode which reduces the power consumption by writing “1” to the standby control register STANDBY. Standby mode is also established after power-on or other system reset operation.

Register List of Standby setting

Register	Register details			Initial value	Setting value	Remarks
	Chip ID	Address ( ): I²C	bit			
STANDBY [0]	02h	00h (3000h)	[0]	1	1: Standby 0: Operating	Register communication is executed even in standby mode.

The serial communication registers hold the previous values. However, the address registers transmitted in standby mode are overwritten. The serial communication block operates even in standby mode, so standby mode can be canceled by setting the STANDBY register to “0”. Some time is required for sensor internal circuit stabilization after standby mode is canceled. For details on the sequence of setting and cancel of standby mode, see the section of “Sensor Setting Flow”.

After standby mode is canceled, a normal image is output from the 9 frames after internal regulator stabilization (6.8 ms or more).



Sequence from Standby Cancel to Stable Image Output

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16.2. Slave Mode and Master Mode

The sensor can be switched between slave mode and master mode.
The switching is made by the XMASTER pin. Establish the XMASTER pin status before canceling the system reset. (Do not switch this pin status during operation.)

Slave Mode (XMASTER = 1h)

Input a vertical sync signal to XVS and input a horizontal sync signal to XHS. For sync signal interval, input data lines to output for vertical sync signal and 1 H period designated in each operating mode for horizontal sync signal. See the section of "Readout Drive Modes" for the number of output data line and 1 H period.
In this sensor, the low-power consumption function is always ON, so be sure to set HMAX [15:0] even in slave mode. The setting value of HMAX [15:0] can be calculated using the following.

HMAX [15:0] = “Number of INCK in 1H” × 74.25 / INCK Frequency

Master Mode (XMASTER = 0h)

Set “XMSTA = 0” in order to start the operation. In addition, set the count number of sync signal in vertical direction by the VMAX [23:0] register and the clock number in horizontal direction by the HMAX [15:0] register. See the description of operation mode for details of the section of “Readout Drive Modes”.
In master mode, it is possible to output horizontal / vertical sync signals from XHS pin / XVS pin by setting register: SYNCSEL [1:0].

Pin Processing

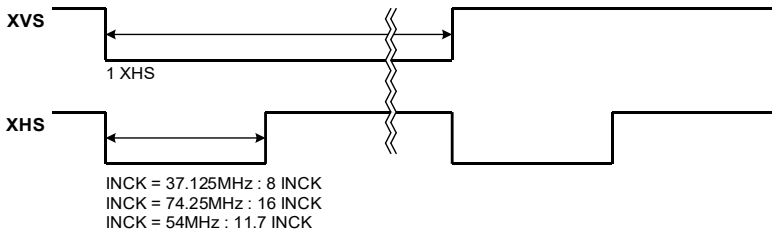
Pin name	Pin processing	Operation mode	Remarks
XMASTER pin	Low fixed	Master mode	High: OV_{DD} Low: GND
	High fixed	Slave mode	

Register List of Slave Mode and Master Mode

Register	Register details			Initial value	Setting value	Remarks
	Chip ID	Address ( ): I^2C	Bit			
XMSTA [0]	02h	10h (3010h)	[0]	1	1: Master operation ready (Initial value) 0: Master operation start	The master operation starts by setting 0.
VMAX [23:0]		D4h (30D4h)	[7:0]	0008AAh	See the item of each drive mode	Line number per frame designated (Master mode and Slave mode common setting.)
		D5h (30D5h)	[7:0]			
		D6h (30D6h)	[7:0]			
HMAX [15:0]		D8h (30D8h)	[7:0]	0167h	See the item of each drive mode	Clock number per line designated (Master mode and Slave mode common setting.)
		D9h (30D9h)	[7:0]			
SYNCSEL [1:0]	06h	3Ch (343Ch)	[5:4]	0h	0h: Normal Output 3h: Hi-Z	XHS, XVS pin setting in Master mode

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XVS / XHS Output in Master Mode



The XVS and XHS are output in timing that set “XMSTA = 0” after standby is released. The XVS and XHS are output asynchronous with other input or output signals. In addition, the output signals are output with an undefined latency time (system delay) relative to the XHS. Therefore, refer to the packet headers output from the sensor and perform synchronization.

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16.3. Gain Adjustment Function

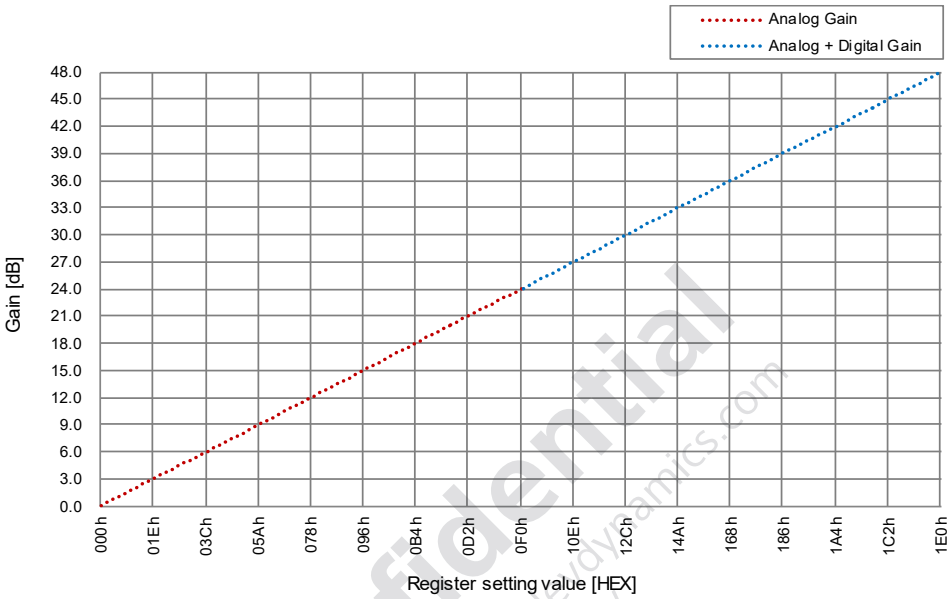
PGC

The Programmable Gain Control (PGC) of this device consists of the analog block and digital block. The total of analog gain and digital gain can be set up to 48 dB by the GAIN [8:0] register setting.

The value which is ten times the gain is set to register. (0.1 dB step)

Example)
When set to 6 dB:

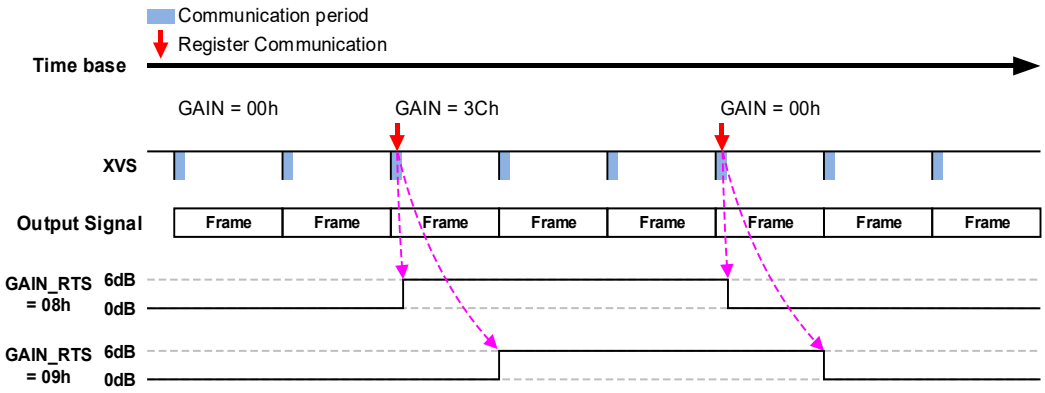
6 × 10 = 60d, GAIN = 03Ch



Register List of Gain setting

Register	Register details			Initial value	Setting value	Remarks
	Chip ID	Address ( ): I²C	bit		Setting range	
GAIN_RTS [7:0]	07h	02h (3502h)	[7:0]	00h	08h or 09h	08h: Gain reflect at the frame 09h: Gain reflect at the next frame
GAIN [8:0]		14h (3514h)	[7:0]	000h	000h to 1E0h (0d to 480d)	Setting value: Gain [dB] × 10
		15h (3515h)	[0]			

Gain Reflection Timing is changed by the set value of the GAIN_RTS [7:0] register as shown below.



Gain Reflection Timing

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16.4. Black Level Adjustment Function

The black level offset (offset variable range: 000h to FFFh) can be added relative to the data in which the digital gain modulation was performed by the BLKLEVEL [11:0] register. When the BLKLEVEL [11:0] setting is increased by 1 LSB, the black level is increased by 1 LSB.

* Use with values shown below is recommended.
  8 bit output: 00Fh (15d)
  10 bit output: 03Ch (60d)
  12 bit output: 0F0h (240d)

Register List of Black level adjustment

Register	Register details			Initial value	Setting value
	Chip ID	Address ( ): I²C	bit		
BLKLEVEL [11:0]	07h	B4h (35B4h)	[7:0]	03Ch	000h to FFFh
		B5h (35B5h)	[3:0]		

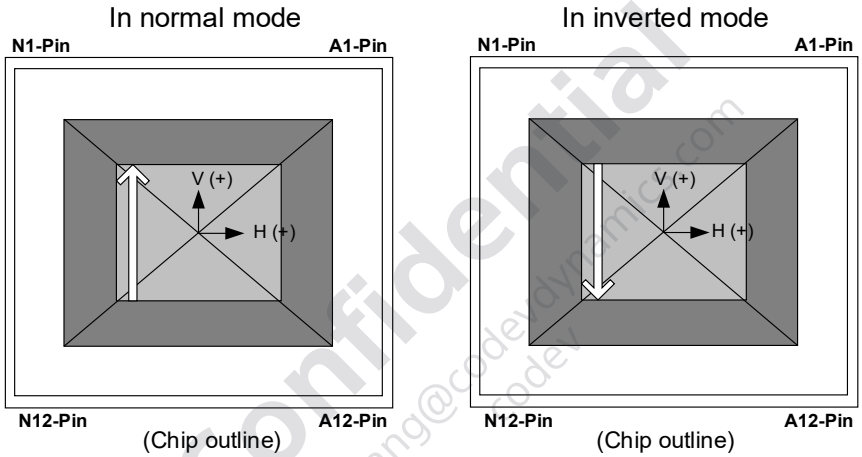
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16.5. Horizontal / Vertical Normal Operation and Inverted Operation

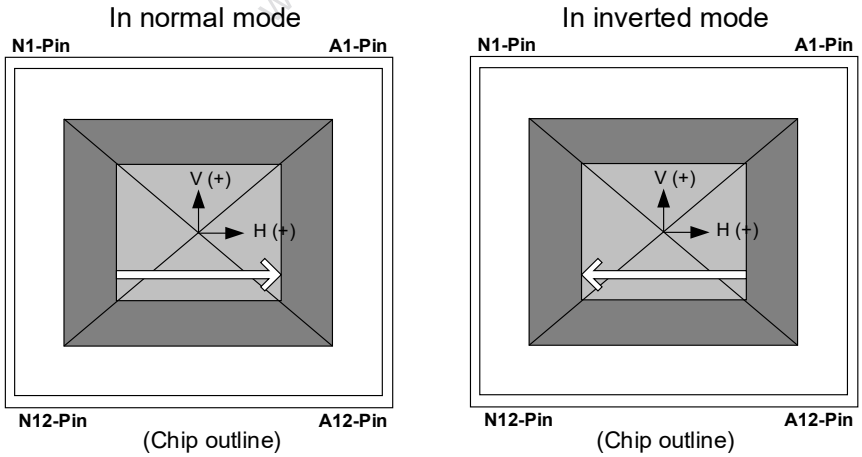
The sensor readout direction (normal / inverted) in vertical direction can be switched by the VREVERSE register setting and sensor readout direction (normal / inverted) in horizontal direction can be switched by the HREVERSE register setting. See the section of “Readout Drive Modes” for the order of readout lines in normal and inverted modes.
One invalid frame is generated when reading immediately after the readout vertical direction change in order to switch the normal operation and inversion between frames.

Register List of Readout Drive Direction setting

Register	Register details			Initial value	Setting value
	Chip ID	Address ( ): I²C	bit		
VREVERSE [0]	04h	04h (3204h)	[0]	0	0: Normal (Initial value) 1: Inverted
HREVERSE [0]			[1]	0	0: Normal (Initial value) 1: Inverted



Normal and Inverted Drive Outline in Vertical Direction (TOP VIEW)



Normal and Inverted Drive Outline in Horizontal Direction (TOP VIEW)

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16.6. Shutter and Exposure Time Settings

This sensor has a global shutter function that integrates to all line collectively by using memory in each pixel. This sensor has a variable electronic shutter function that can control the exposure time in line units for adjust the exposure time. This sensor transferred signal to memory in pixel after the exposure (memory transfer), then this sensor performs output in which readout operation is performed sequentially for each line in sync with the XHS signal. This sensor has trigger mode that can be controlled exposure start timing and memory transfer timing by trigger.

Note) For exposure time control, an image which reflects the setting is output from the frame after the setting changes.

In this item, the shutter operation and storage time are shown as in the figure below with the time sequence on the horizontal axis and the vertical address on the vertical axis. For simplification, shutter and readout operation are noted in line units.

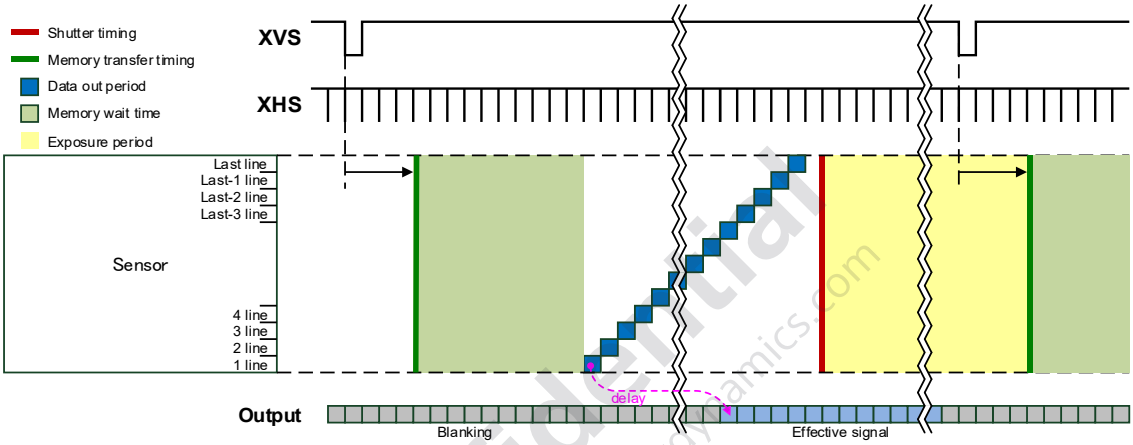


Image Drawing of Global Shutter (Normal mode) Operation

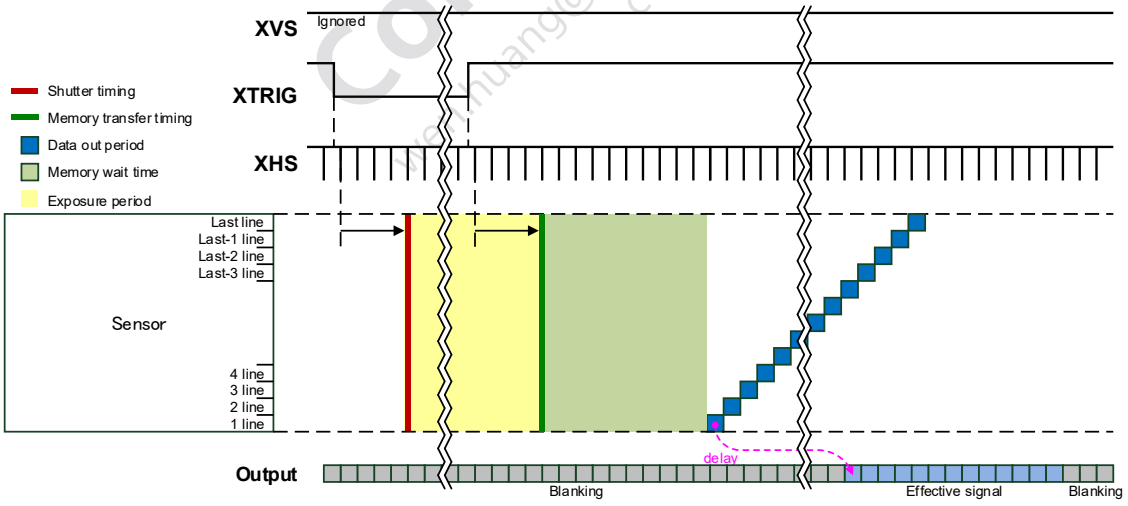


Image Drawing of Global Shutter (Sequential Trigger mode) Operation

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16.6.1. Global Shutter (Normal Mode) Operation

The exposure time can be controlled by varying the electronic shutter timing. In the electronic shutter settings, the exposure time is controlled by the SHS [23:0] register. For setting value of SHS [23:0], see the table “List of Exposure Setting”. When the sensor is operating in slave mode, the number of lines per frame is determined by the XVS interval (number of lines), using the input XHS interval as the line unit. When the sensor is operating in master mode, the number of lines per frame is determined by the VMAX [23:0] register. The number of lines per frame differs according to the readout mode.
When XVS is input at intervals shorter than the recommended setting value of VMAX [23: 0] in each readout mode, interrupt operation is performed.

Calculation Formula of Exposure Time

Exposure time [µs] = (1 H period [µs]) × (Number of lines per frame - SHS) + 2.47 [µs][*1]
[*1]: Exposure time error (t_{OFFSET})

Register List of Shutter setting

Register	Register details			Initial value	Setting value
	Chip ID	Address ( ): I²C	bit		
VMAX [23:0]	02h	D4h (30D4h)	[7:0]	0008AAh	Set the number of lines per frame (only in master mode)
		D5h (30D5h)	[7:0]		
		D6h (30D6h)	[7:0]		
GMRWT [7:0]		E2h (30E2h)	[7:0]	06h	For setting, refer to the register list of each readout mode
GMTWT [7:0]		E3h (30E3h)	[7:0]	28h	For setting, refer to the register list of each readout mode
GSDLY [7:0]		E6h (30E6h)	[7:0]	12h	For setting, refer to the register list of each readout mode
VINT_EN_NOR [0]	04h	3Eh (323Eh)	[1]	1	Setting of Interrupt mode in Normal Mode 0: V interrupt is disable 1: V interrupt is enable
SHS [23:0]		40h (3240h)	[7:0]	000070h	Sets the shutter sweep time. (memory wait time (GMTWT value) + 4) to (Number of lines per frame - 1)
		41h (3241h)	[7:0]		
		42h (3242h)	[7:0]		

When set VMAX [23:0] to a number of V lines or more than that noted for each operating mode, the imaging characteristics are not guaranteed.

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List of Exposure Setting

Drive mode	memory transfer stabilization wait time [H]	Number of lines per frame [DEC]	SHS Setting value [DEC]	Exposure Setting value [H]	AD 12 bit, 1188 Mbps/lane, 4 lane output (Maximum frame rate)	
					Frame rate [frame/s]	Actually exposure [ms][*3]
All-pixel	30 (GMTWT[*1])	2198 (VMAX)	2197	1	67.2	0.009
			2196	2		0.016
			…	…		…
			35	2163		14.626
			34	2164		14.633
1/2 Subsampling, 2 × 2 FD binning	56 (GMTWT[*1])	1176 (VMAX)	1175	1	232.9	0.006
			1174	2		0.010
			…	…		…
			61	1115		4.072
			60	1116		4.076
ROI Overlap ROI	All-pixel (GMTWT[*1])	V_{TR}[*2]	V_{TR}-1	1	[*2]	0.009
			V_{TR}-2	2		0.016
			…	…		…
			35	V_{TR}-35		[*4]
			34	V_{TR}-34		

[*1] For GMTWT, refer to the register list of each readout mode.
[*2] V_{TR} and the frame rate, refer to the section “ROI mode” in “Readout Drive Modes”.
[*3] INCK frequency is input by typical value. and t_{OFFSET} (2.47 [μs]) is included.
[*4] Conform to the calculation formula of exposure time. (Number of lines per frame = V_{TR})

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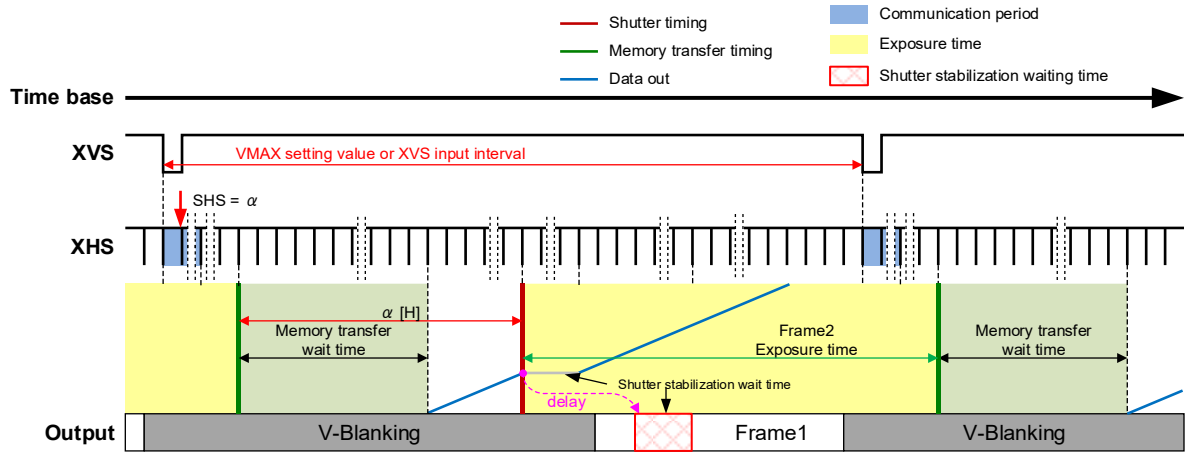


Image Drawing of Global Shutter (Normal Mode)

Interrupt Operation

In case of “VINT_EN_NOR = 1”, the image drawing when the interrupt operation is generated is shown below. When the next XVS is input during read of the frame (Frame 1 in the figure below), Frame 1 becomes an invalid frame. XVS input timing of interrupt generating corresponds to recommended setting value of VMAX [23:0] in each readout mode.
In case of "VINT_EN_NOR = 0", XVS input at intervals shorter than the recommended setting value of VMAX [23:0] is invalid.

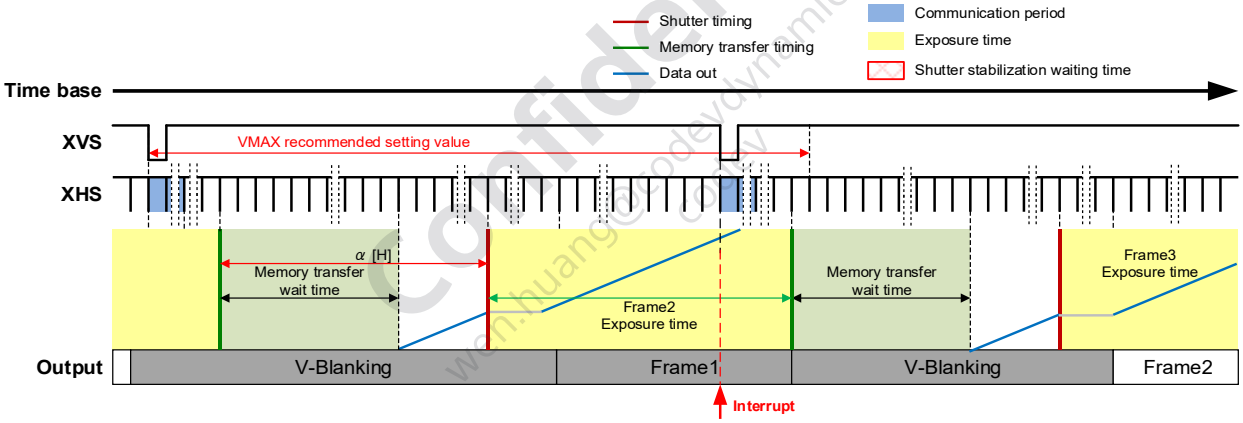


Image Drawing of Interrupt Operation in Global Shutter (Normal Mode)

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16.6.2. Global Shutter (Sequential Trigger Mode) Operation

The exposure time can be controlled by varying the pulse width that is input to XTRIG1 pin. The pulse width designated in XHS unit [H]. For the transition from normal mode to Sequential trigger mode, set 1 to the register TRIGMODE [2:0]. The XVS input signal is ignored during Sequential trigger mode operating. In case of inputting trigger continuously, there are period which prohibit the trigger rise input (t_{TGPD}) and fall input (t_{TGES}) based on the previous trigger rise.
When the trigger rise is input before the rise input prohibited period (t_{TGPD}), interrupt operation starts.
This function is slave mode only. The number of lines per frame differs according to the operating mode.
XTRIG2 pin is set Open (Hi-Z) or fixed to High after power-on sequence.

Calculation Formula of Exposure Time

Exposure time [µs] = (1 H period [µs]) × (XTRIG low level pulse width [H][*2]) + 2.47 [µs][*1]
[*1]: Exposure time error (t_{OFFSET})
[*2]: Low level pulse width is counted by XHS pulse.

Register List of shutter setting

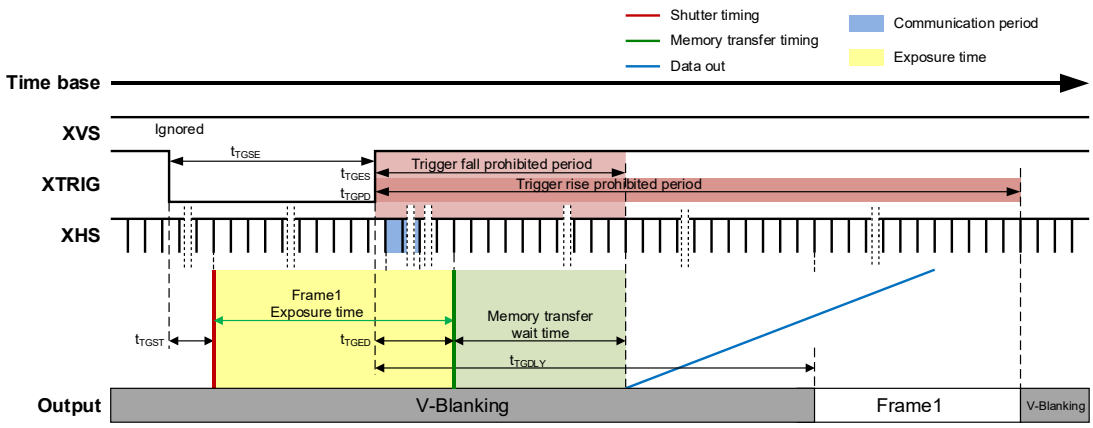
Register	Register details			Initial value	Setting value
	Chip ID	Address ( ): I²C	bit		
XMSTA [0]	02h	10h (3010h)	[0]	1	Setting of master mode operation 0: Master mode operation start 1: Master mode operation stop
VINT_EN [0]	04h	3Eh (323Eh)	[0]	1	Setting of Interrupt mode in Trigger Mode 0: V interrupt is disable 1: V interrupt is enable
TRIGMODE [2:0]	06h	00h (3400h)	[2:0]	0h	Global shutter mode setting 0: Normal mode 1: Sequential Trigger mode 2: Fast Trigger mode
TRIGTIMING [1:0]	06h	00h (3400h)	[4:3]	0h	Trigger mode setting 0h: Normal mode 1h: Trigger mode Others: Setting prohibited

Parameter List of Global Shutter (Sequential Trigger Mode)

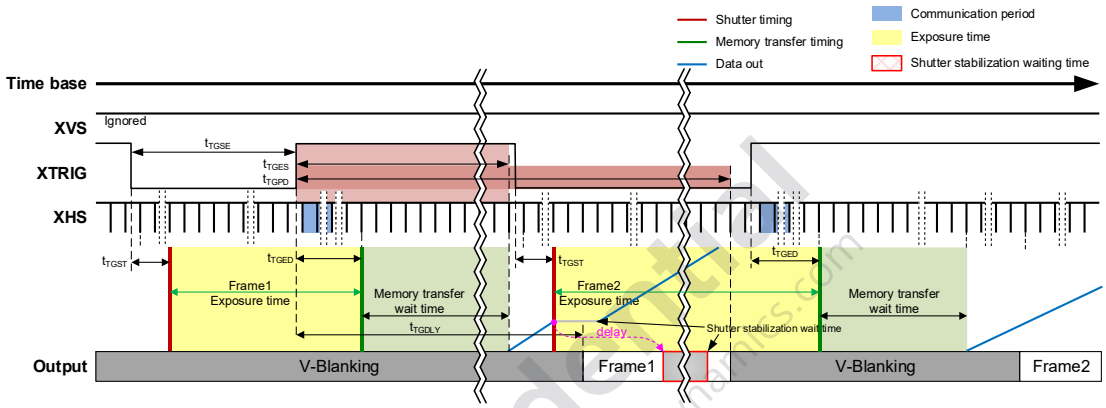
Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay (All-pixel, ROI, 1/2 Subsampling, 2 × 2 FD binning)	t_{TGST}	4 + GMRWT[*1]	—	5 + GMRWT[*1]	H
Integration end delay (All-pixel, ROI, 1/2 Subsampling, 2 × 2 FD binning)	t_{TGED}	4 + GMRWT[*1] + t_{OFFSET}	—	5 + GMRWT[*1] + t_{OFFSET}	H
Pulse width	t_{TGSE}	1	—	—	H
Next trigger fall prohibited period (All-pixel, ROI, 1/2 Subsampling, 2 × 2 FD binning)	t_{TGES}	5 + GMRWT[*1] + GMTWT[*1]	—	—	H
Next trigger rise prohibited period (All-pixel, 1/2 Subsampling, 2 × 2 FD binning)	t_{TGPD}	VMAX	—	—	H
Next trigger rise prohibited period (ROI)		V_{TR}[*2]	—	—	
Data output delay (All-pixel / ROI)	t_{TGDLY}	—	15 + GMRWT[*1] + GMTWT[*1]	—	H
Data output delay (1/2 Subsampling / 2 × 2 FD binning)			13 + GMRWT[*1] + GMTWT[*1]		

[*1] For GMRWT, GMTWT, refer to the register list of each readout mode.
[*2] V_{TR} (See the section “ROI mode” in “Readout Drive Modes”)

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Single shutter Image Drawing of Global Shutter (Sequential Trigger Mode)



Multi shutter image Drawing of Global Shutter (Sequential Trigger Mode)

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Interrupt Operation

In case of “VINT_EN = 1”, the image drawing when the interrupt operation is generated is shown below. When the trigger is raised again and the next frame is output during read of the frame for which read was started by a trigger rise (Frame 1 in the figure below), Frame 1 becomes an invalid frame. Trigger timing of interrupt generating corresponds to t_{TGPD} in Parameter List of Global Shutter (Trigger Mode)
In case of “VINT_EN = 0”, both of the rising edge and the falling edge of the trigger signal are ignored in t_{TGPD} (Prohibit period).

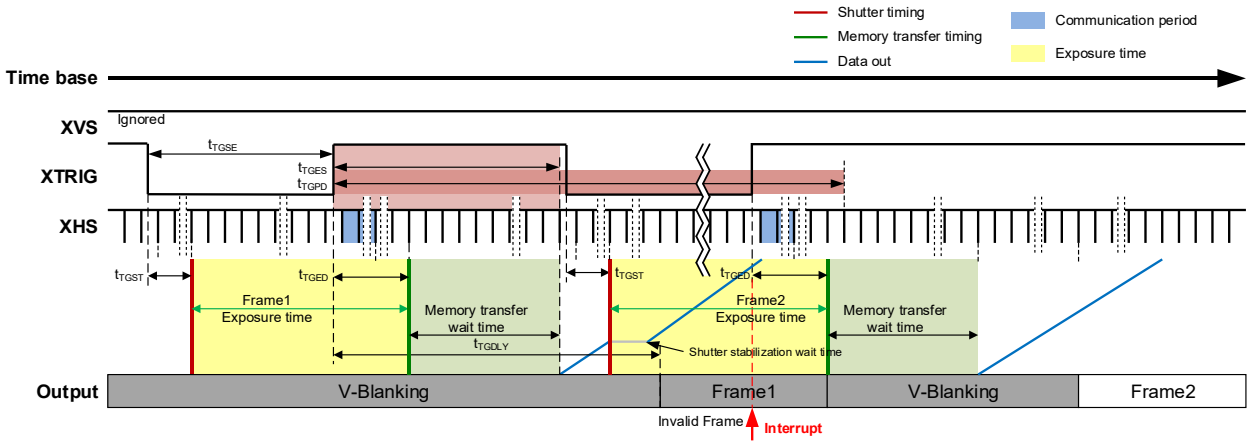


Image Drawing of Interrupt Operation in Global Shutter (Sequential Trigger Mode)

16.6.3. Global Shutter (Fast Trigger Mode) Operation

Fast trigger mode is the trigger mode that starts exposure at fall of XTRIG1 immediately.
In the Fast trigger mode, set the register TRIGMODE [2:0] to 2h at standby state, and then release standby. In the Fast trigger mode, the image size can not be changed during the operation.
This mode supports Master mode only.
XTRIG2 pin is set Open (Hi-Z) or fixed to High after power-on sequence.

Calculation Formula of Exposure Time

Exposure time [µs] = (XTRIG low level pulse width [µs]) + (GMRWT [H] × 1 H period [µs]) + 2.53 [µs][*1]
[*1]: Exposure time error (t_{OFFSET})

Register List of shutter setting

Register	Register details			Initial value	Setting value
	Chip ID	Address ( ): I²C	bit		
XMSTA [0]	02h	10h (3010h)	[0]	1	Setting of master mode operation 0: Master mode operation start 1: Master mode operation stop
TRIGMODE [2:0]	06h	00h (3400h)	[2:0]	0h	Global shutter mode setting 0: Normal mode 1: Sequential Trigger mode 2: Fast Trigger mode
TRIGTIMING [1:0]	06h	00h (3400h)	[4:3]	0h	Trigger mode setting 0h: Normal mode 1h: Trigger mode Others: Setting prohibited

Parameter List of Global Shutter (Fast Trigger Mode)

Item	Symbol	Min.	Typ.	Max.	Unit
Integration start delay (All-pixel, ROI, 1/2 Subsampling, 2 × 2 FD binning)	t_{TGST}	—	—	0.1	µs
Integration end delay (All-pixel, ROI, 1/2 Subsampling, 2 × 2 FD binning)	t_{TGED}	—	—	0.1 + GMRWT[*1] + t_{OFFSET}	µs
Pulse width	t_{TGSE}	0.05	—	—	µs
Next trigger rise / fall prohibited period (All-pixel, 1/2 Subsampling, 2 × 2 FD binning)	t_{TGPD}	VMAX	—	—	H
Next trigger rise / fall prohibited period (ROI)		V_{TR}[*2]	—	—	
Data output delay (All-pixel / ROI)	t_{TGDLY}	—	11 + GMRWT[*1] + GMTWT[*1]	—	H
Data output delay (1/2 Subsampling / 2 × 2 FD binning)			9 + GMRWT[*1] + GMTWT[*1]		

[*1] For GMRWT, GMTWT, refer to the register list of each readout mode.
[*2] V_{TR} (see the section “ROI mode” in “Readout Drive Modes”).

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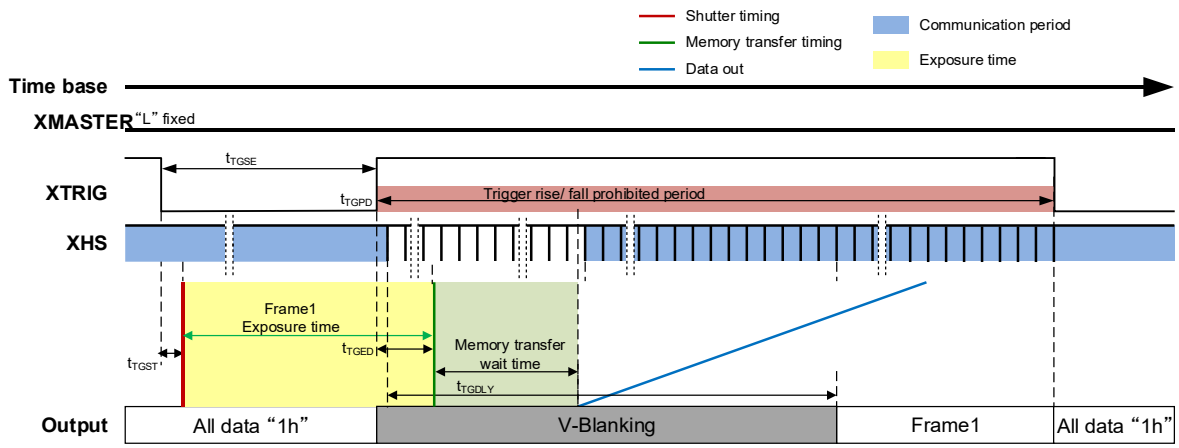


Image Drawing of Global Shutter (Fast Trigger Mode) (4-wire)

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16.6.4. Mode Transitions of Global Shutter Operation

The sensor can be switched between normal mode and trigger mode in global shutter operation by setting the register TRIGMODE [2:0]. The sensor will transition to normal mode or trigger mode “GMRWT [7:0] (H) + GMTWT [7:0] (H)” after the register TRIGMODE [2:0] is set.
(The XVS and XTRIG input during transition are prohibited.)
In case of Fast Trigger mode, the mode transition must be done via sensor standby.

Transition from Normal Mode to Sequential Trigger Mode

The sensor will transition from normal mode to Sequential trigger mode after setting “TRIGMODE [2:0] = 1h”. The XVS input is ignored after transition to trigger mode. Trigger input is prohibited for a “GMRWT [7:0] (H) + GMTWT [7:0] (H)” period after the register TRIGMODE [2:0] is set. When TRIGMODE [2:0] is set during data read, read operation is stopped and that frame becomes an invalid frame.

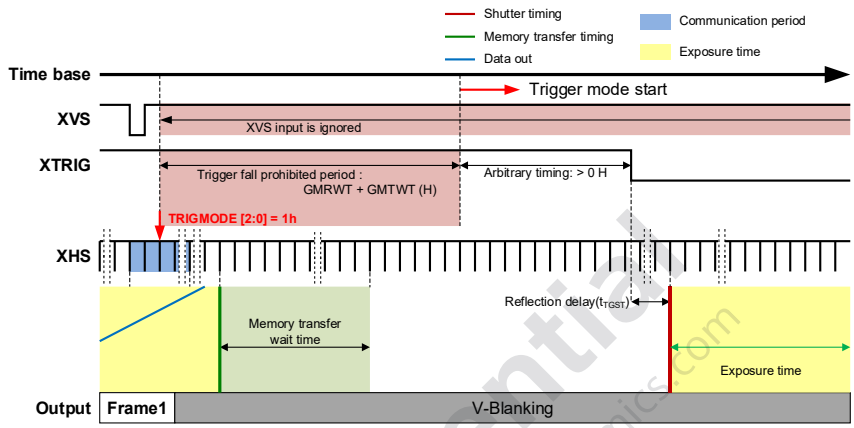


Image Drawing of Transition from Normal Mode to Sequential Trigger Mode

Transition from Sequential Trigger Mode to Normal Mode

The sensor will transition from trigger mode to normal mode after setting “TRIGMODE [2:0] = 0h”. Start XVS input after transition to normal mode. Set TRIGMODE [2:0] after Next trigger rise prohibited period (t_{TGPD}) has passed. When TRIGMODE [2:0] is set before t_{TGPD}, read operation is stopped and that frame becomes an invalid frame.

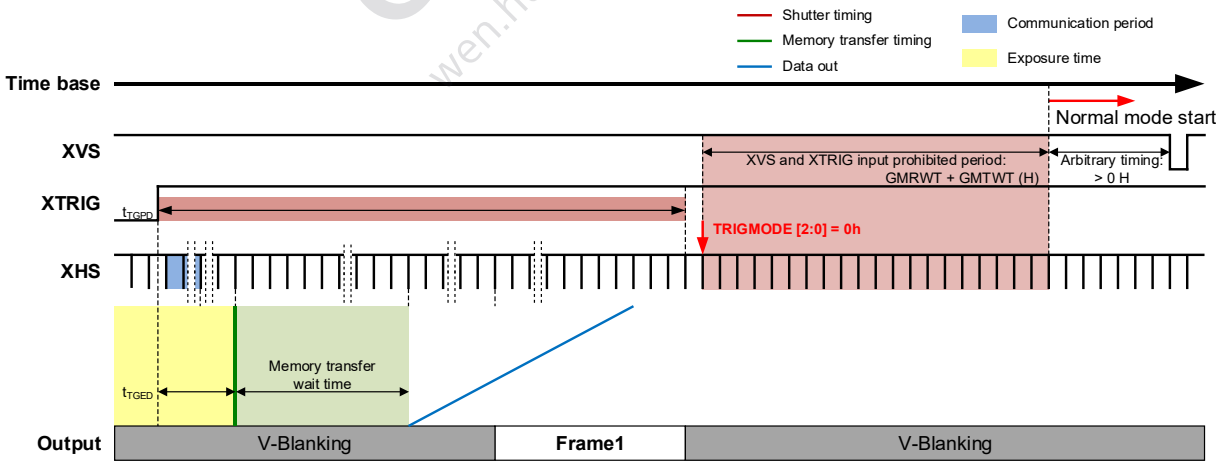


Image Drawing of Transition from Sequential Trigger Mode to Normal Mode

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16.6.5. Pulse Output Function

This sensor has a pulse output function that indicates each state of shutter operation.

TOUT0 pin: The monitor output for Exposure period
TOUT1 pin: Programmable pulse
TOUT2 pin: Programmable pulse

(1) Exposure period control signal

This function is to output the actual exposure period control signal from the TOUT0 pin by setting the following registers.
This function can also be used in Fast trigger mode.

Register List of Pulse Output Function

Register	Register details			Initial value	Setting value
	Chip ID	Address ( ): I²C	bit		
TOUT0SEL [2:0]	06h	36h (3436h)	[2:0]	0h	TOUT0 pin setting 0h: Low fixed 1h: The monitor output for Exposure period

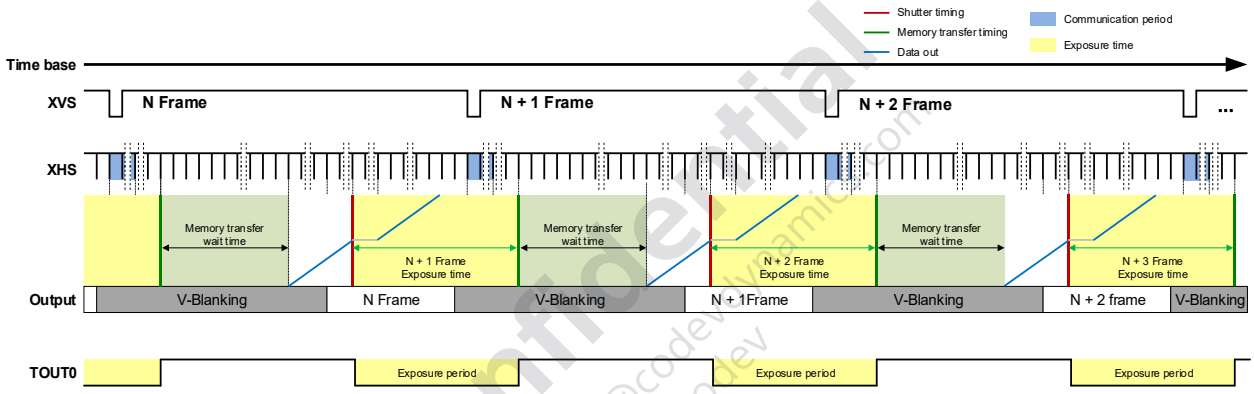


Image Drawing of the monitor output for Exposure period in Global Shutter (Normal Mode)

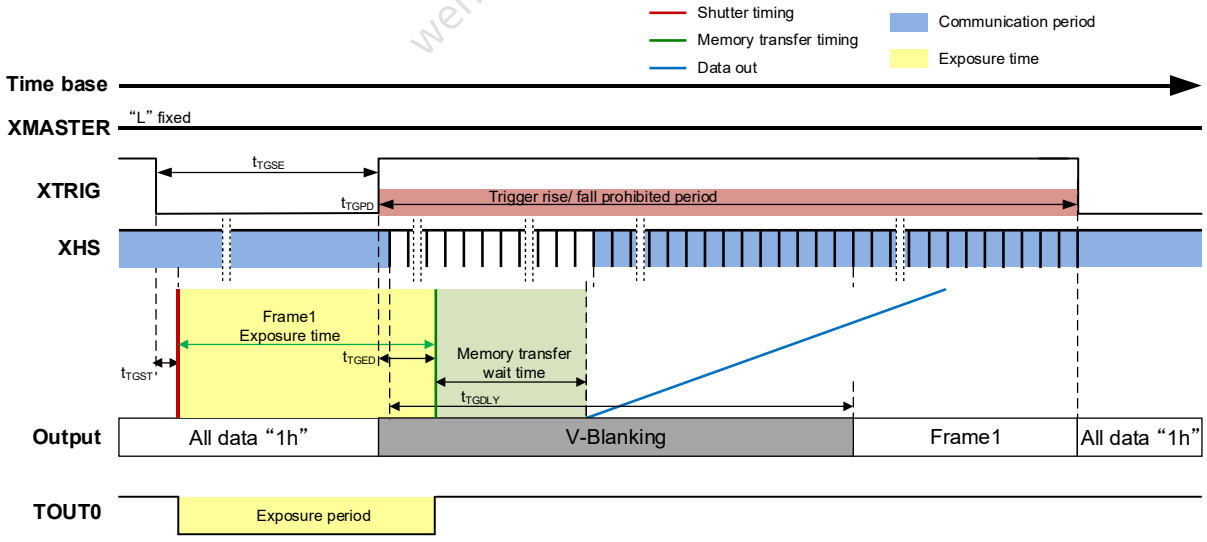


Image Drawing of the monitor output for Exposure period in Global Shutter (Fast Trigger Mode)

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(2) Programmable Pulse signal

This function outputs programmable pulses from the TOUT1 and TOUT2 pins. The rise timing and fall timing of programmable pulse are set by registers, so user can generate the flexible pulses. For the reference point (The timing when register value set to 0) to be set, see the table “List of Reference point". The programmable pulse is output asynchronously with other signals on the basis of the sensor internal timing shown in the table “List of Reference point".
This function doesn't support Fast Trigger mode.

Register List of Pulse Output Function

Register	Register details			Initial value	Setting value
	Chip ID	Address ( ): I²C	bit		
TOUT1SEL [1:0]	06h	35h (3435h)	[1:0]	0h	TOUT1 pin setting 0h: Low fixed 3h: Pulse output
TOUT2SEL [1:0]			[3:2]	0h	TOUT2 pin setting 0h: Low fixed 3h: Pulse output
TRIG_TOUT1_SEL [3:0]		3Ah (343Ah)	[3:0]	0h	TOUT1 pin output selection 0h: Low fixed 1h: Pulse1 output
TRIG_TOUT2_SEL [3:0]			[7:4]	0h	TOUT2 pin output selection 0h: Low fixed 2h: Pulse2 output
PULSE1_EN_NOR [0]		78h (3478h)	[0]	0	Pulse1 enable in normal mode 0: disable 1: enable
PULSE1_EN_TRIG [0]			[1]	0	Pulse1 enable in trigger mode 0: disable 1: enable
PULSE1_POL [0]			[2]	0	Pulse1 polarity selection 0: High active 1: Low active
PULSE1_UP [23:0]		79h (3479h)	[7:0]	000000h	Pulse1 active period start timing setting Designated in line units from reference point
		7Ah (347Ah)	[7:0]		
		7Bh (347Bh)	[7:0]		
PULSE1_DN [23:0]		7Ch (347Ch)	[7:0]	000000h	Pulse1 active period end timing setting Designated in line units from reference point
		7Dh (347Dh)	[7:0]		
		7Eh (347Eh)	[7:0]		
PULSE2_EN_NOR [0]		80h (3480h)	[0]	0	Pulse2 enable in normal mode 0: disable 1: enable
PULSE2_EN_TRIG [0]			[1]	0	Pulse2 enable in trigger mode 0: disable 1: enable
PULSE2_POL [0]			[2]	0	Pulse2 polarity selection 0: High active 1: Low active
			[5]	0	Fixed to 1
PULSE2_UP [23:0]		81h (3481h)	[7:0]	000000h	Pulse2 active period start timing setting Designated in line units from reference point
		82h (3482h)	[7:0]		
		83h (3483h)	[7:0]		
PULSE2_DN [23:0]		84h (3484h)	[7:0]	000000h	Pulse2 active period end timing setting Designated in line units from reference point
		85h (3485h)	[7:0]		
		86h (3486h)	[7:0]		

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List of Reference Point

	Normal mode	Trigger mode
Reference point of Pulse1	XVS fall edge in N frame	Fall edge of input trigger
Reference point of Pulse2	XVS fall edge in N +1 frame	Rise edge of input trigger

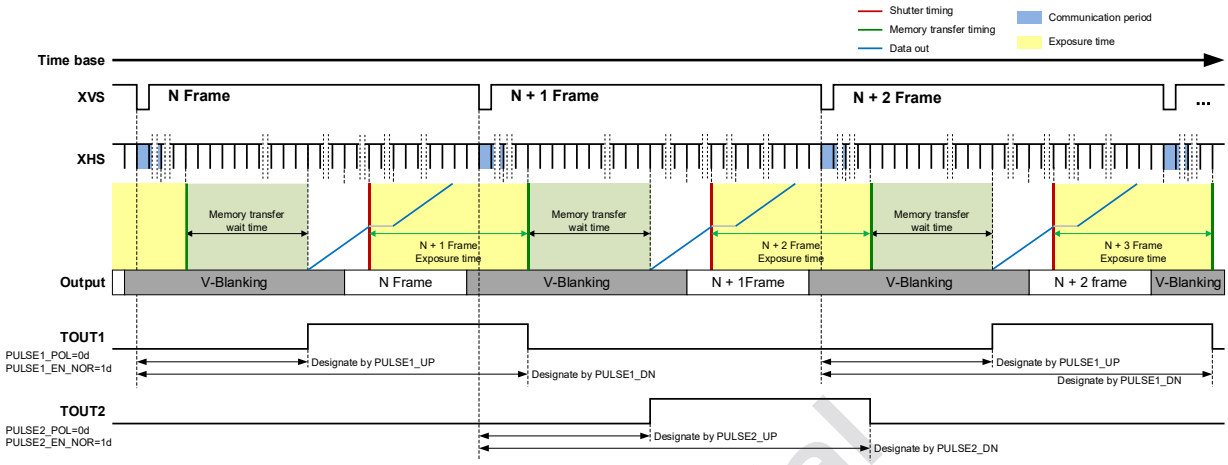


Image Drawing of Pulse Output Function in Global Shutter (Normal Mode)

In normal mode, TOUT1 and TOUT2 are output alternately each time inputting XVS.

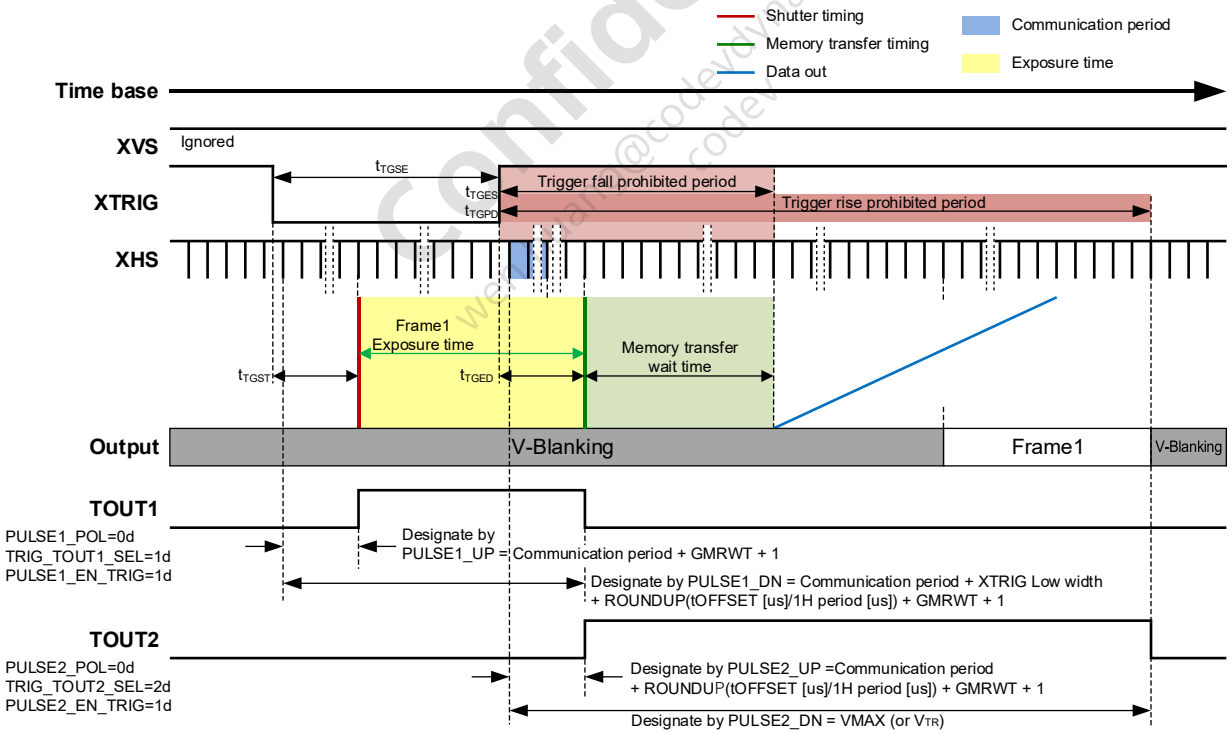


Image Drawing of Pulse Output Function in Global Shutter (Sequential Trigger Mode)

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16.7. Signal Output

16.7.1. Output Pin Settings

The output formats of this sensor support the following modes.

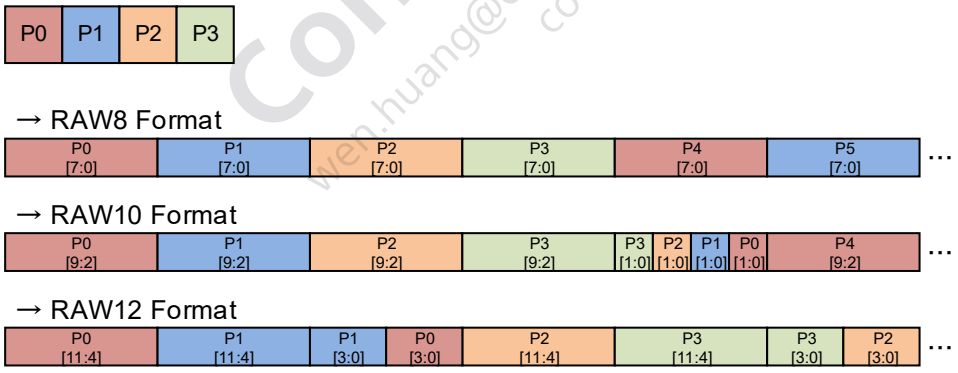
CSI-2 serial 2 Lane / 4 Lane, RAW8 / RAW10 / RAW12

The 2 Lane / 4 Lane serial signal output method using this sensor is described below.
Complied with the CSI-2, data is output using 2 Lane / 4 Lane. The image data is output from the CSI-2 output pin.
The DOP0 / DOM0 are called the Lane1 data signal, the DOP1 / DOM1 are called the Lane2 data signal, the DOP2 / DOM2 are called the Lane3 data signal, the DOP3 / DOM3 are called the Lane4 data signal. In addition, the clock signals are output from DCKP / DCKM of the CSI-2 pins.
In 2 Lane mode, data is output from Lane1 and Lane2. In 4 Lane mode, data is output from Lane1, Lane2, Lane3 and Lane4.
The value of maximum data rate is 1188 Mbps / Lane.
The select of RAW8, RAW10 or RAW12 is set by the register: ODBIT [1:0]. The number of output lanes is set by the register: LANESEL [2:0].
Unused lanes output signals conformed to MIPI standard.

Register List of Output Settings

Register	Register details			Initial value	Setting value
	Chip ID	Address ( ): I²C	bit		
ODBIT [1:0]	06h	30h (3430h)	[1:0]	0h	0h: 10 bit 1h: 12 bit 2h: 8 bit
LANESEL [2:0]	0Bh	04h (3904h)	[2:0]	0h	CSI-2 IF Output Lane selection 2h: 4 Lane 3h: 2 Lane

The formats of RAW8, RAW10 and RAW12 are shown below.

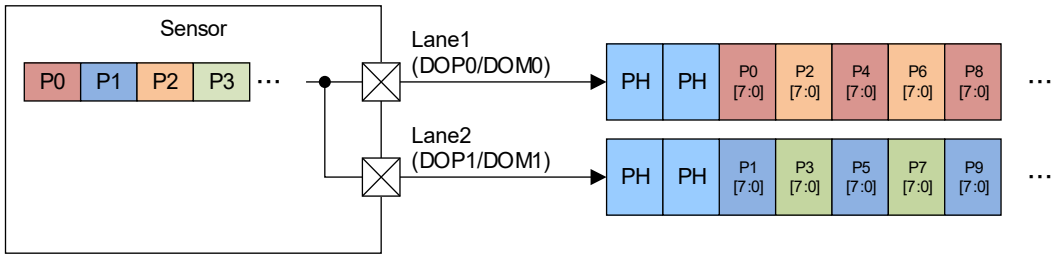


The Example of Format of RAW8 / RAW10 / RAW12

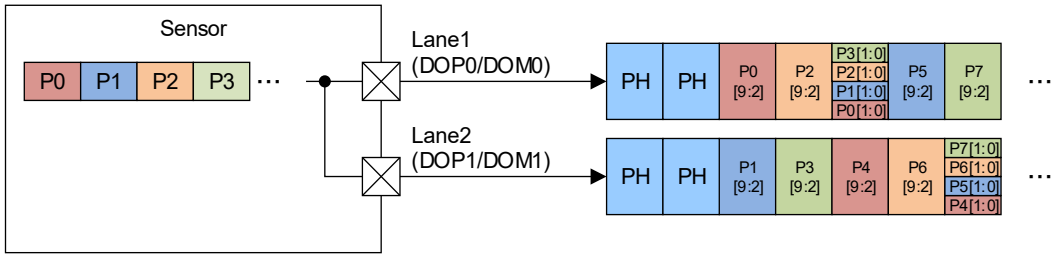
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The each formal of 2 Lane and 4 Lane are shown below.

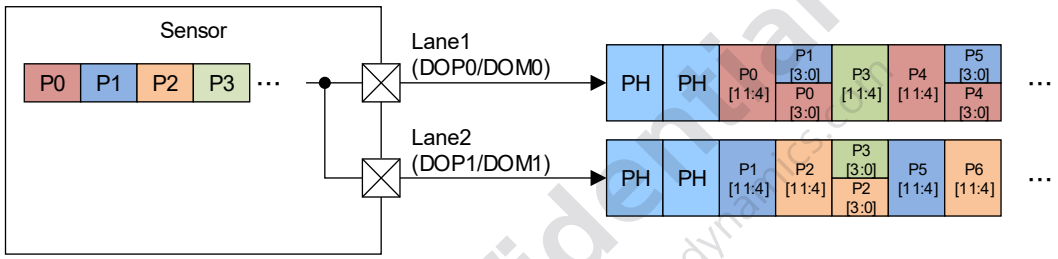
a) 2 Lane-RAW8



b) 2 Lane-RAW10



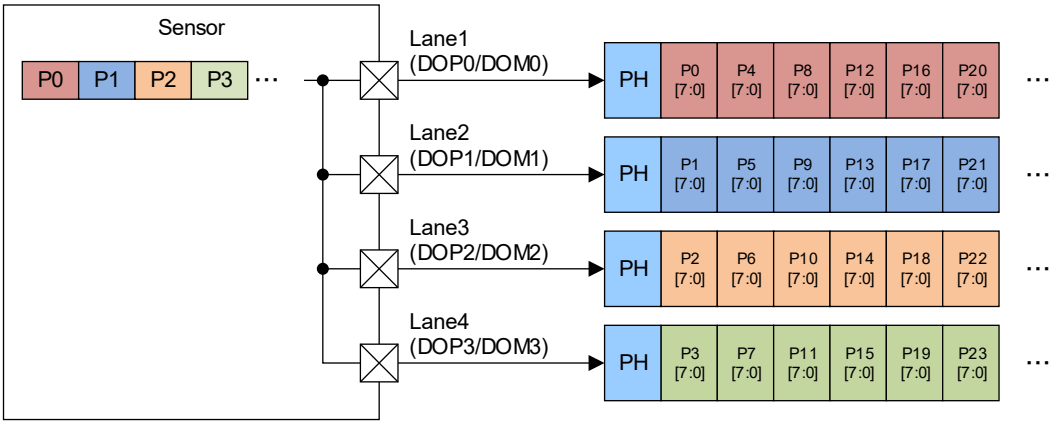
C) 2 Lane-RAW12



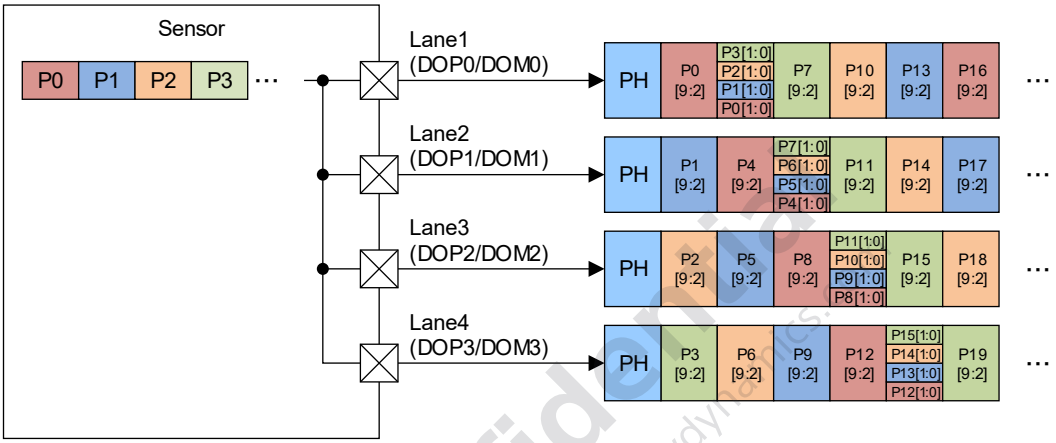
2 Lane Output Format

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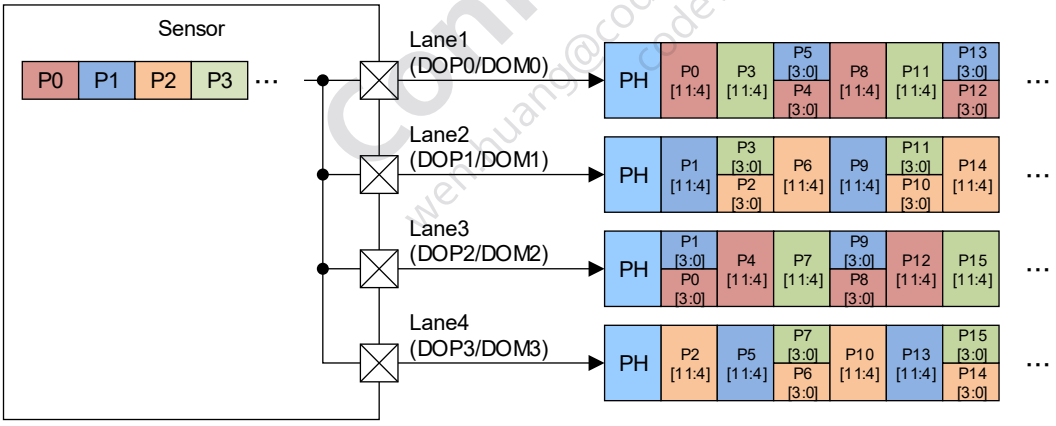
d) 4 Lane-RAW8



e) 4 Lane-RAW10



f) 4 Lane-RAW12

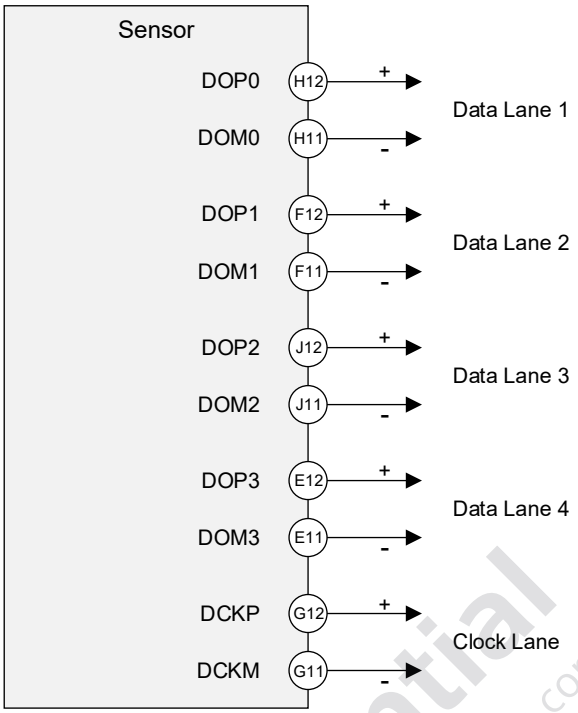


4 Lane Output Format

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16.7.2. MIPI Transmitter

Output pins (DOP0, DOM0, DOP1, DOM1, DOP2, DOM2, DOP3, DOM3, DCKP, DCKM) are described in this section.

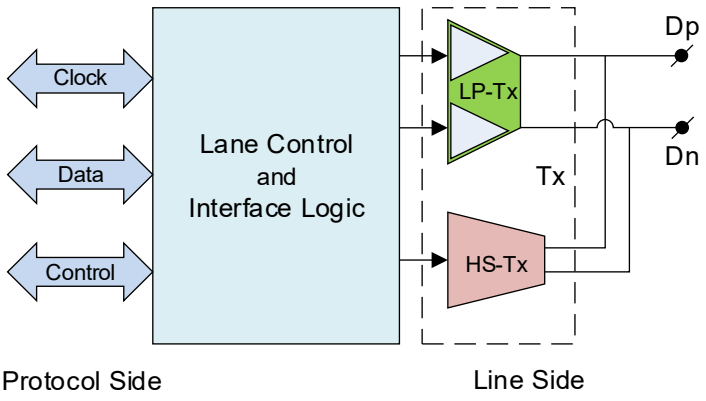


Relationship between Pin Name and MIPI Output Lane

The pixel signals are output by the CSI-2 High-speed serial interface.
See the MIPI Standard

- MIPI Alliance Specification for Camera Serial Interface 2 (CSI-2) Version 1.2
- MIPI Alliance Specification for D-PHY Version 1.2

The CSI-2 transfers one bit with a pair of differential signals. The transmitter outputs differential current signal after converting pixel signals to it. Insert external resistance in differential pair in a series or use cells with a built-in resistance on the Receiver side. When inserting an external resistor, as close as possible to the Receiver. The differential signals maintain a constant interval and reach the receiver with the shortest wiring length possible to avoid malfunction. The maximum bit rate of each Lane is 1188 Mbps / Lane.



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16.7.3. Output Pin Bit Width Selection

The number of internal A/D conversion bits can be selected from 8 bit, 10 bit or 12 bit by the register ADBIT [1:0]. See the section of “Readout Drive Modes” for the correspondence with each mode.

Register List of Bit Width Selection

Register	Register details			Initial value	Setting value	Remarks
	Chip ID	Address ( ) : I^2C	bit			
ADBIT [1:0]	04h	00h (3200h)	[5:4]	0h	0h: 10 bit 1h: 12 bit 2h: 8 bit	Set same value to both ADBIT and ODBIT
ODBIT [1:0]	06h	30h (3430h)	[1:0]	0h	0h: 10 bit 1h: 12 bit 2h: 8 bit	

16.7.4. Output Signal Range

In CSI-2 output mode, the sensor output has either a 8 bit, 10 bit or 12 bit gradation, and the maximum output value is the FFh (8 bit output), 3FFh value (10 bit output) and the FFFh one (12 bit output).
The output range for each output gradation is shown in the table below.

Output Gradation and Output Range

Output gradation	Output value	
	Min.	Max.
8 bit	00h	FFh
10 bit	000h	3FFh
12 bit	000h	FFFh

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16.8. Global Timing Register Setting

The setting values of the Global Timing register are shown below.

Global Timing Register Setting

Register	Register details			Initial value	Setting value		
	Chip ID	Address ( ): I²C	bit		1188 Mbps	891 Mbps	594 Mbps
THS_PREPARE [15:0]	0Eh	B4h (3CB4h)	[7:0]	004Fh	004Fh	003Fh	002Fh
		B5h (3CB5h)	[7:0]				
TCLK_POST [15:0]		B6h (3CB6h)	[7:0]	008Fh	008Fh	007Fh	0067h
		B7h (3CB7h)	[7:0]				
THS_TRAIL [15:0]		B8h (3CB8h)	[7:0]	004Fh	004Fh	003Fh	002Fh
		B9h (3CB9h)	[7:0]				
THS_ZERO [15:0]		BAh (3CBAh)	[7:0]	0087h	0087h	006Fh	004Fh
		BBh (3CBBh)	[7:0]				
TCLK_PREPARE [15:0]		BCh (3CBCh)	[7:0]	004Fh	004Fh	0037h	0027h
		BDh (3CBDh)	[7:0]				
TCLK_TRAIL [15:0]		BEh (3CBEh)	[7:0]	0047h	0047h	0037h	0027h
		BFh (3CBFh)	[7:0]				
TLPX [15:0]		C0h (3CC0h)	[7:0]	003Fh	003Fh	002Fh	0027h
		C1h (3CC1h)	[7:0]				
TCLK_ZERO [15:0]		C2h (3CC2h)	[7:0]	0137h	0137h	0F7h	0B7h
		C3h (3CC3h)	[7:0]				
TCLK_PRE [15:0]		C4h (3CC4h)	[7:0]	000Fh	000Fh	000Fh	000Fh
		C5h (3CC5h)	[7:0]				
THS_EXIT [15:0]		C6h (3CC6h)	[7:0]	007Fh	007Fh	005Fh	0047h
		C7h (3CC7h)	[7:0]				

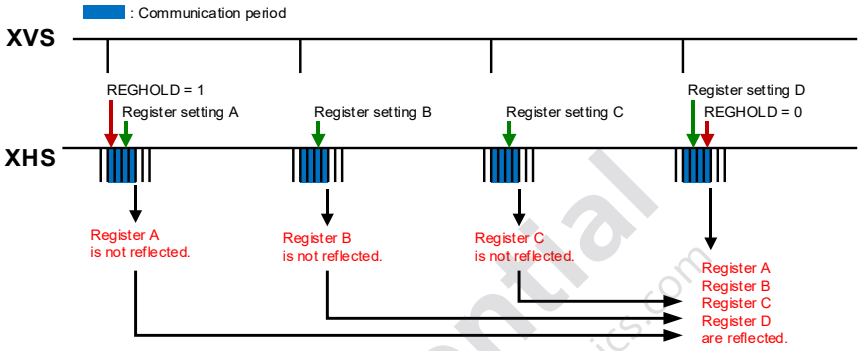
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16.9. Register Hold Setting

Register setting can be transmitted with divided to several frames and it can be reflected globally at a certain frame by the register REGHOLD. Setting REGHOLD = 1 at the start of register communication period prevents the registers that are set thereafter from reflecting at the frame reflection timing. The registers that are set when setting REGHOLD = 1 are reflected globally by setting REGHOLD = 0 at the end of communication period of the desired frame to reflect the register.

Register List of Register Hold

Register	Register details			Initial value	Setting value
	Chip ID	Address ( ): I²C	bit		
REGHOLD [0]	02h	34h (3034h)	[0]	0	0: Invalid 1: Valid (Register hold)



Register Hold Setting

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16.10. Mode Transition

The Mode transition between operations is shown below. These examples shown in case that setting is completed within one communication timing.

List of Mode Transition

Transition			State
Operation mode			
Normal mode (Sensor Slave mode)	→	Sequential Trigger mode	Via the standby state is unnecessary Invalid frame: 1
Sequential Trigger mode	→	Normal mode (Sensor Slave mode)	
Normal mode (Sensor Master mode)	→	Fast Trigger mode	Via the standby state is necessary
Fast Trigger mode	→	Normal mode (Sensor Master mode)	
Readout mode (Normal mode or Sequential Trigger mode)			
All-pixel	→	ROI	Via the standby state is unnecessary Invalid frame: 0
ROI	→	All-pixel	
ROI	→	ROI	
Readout mode (Fast Trigger mode)			
All-pixel	→	ROI	Via the standby state is necessary
ROI	→	All-pixel	
ROI	→	ROI	
- Transition between modes other than the above - Change the input frequency of INCK[*1] - Change the register setting noted “S” in the reflection timing column of the Register Map			Via the standby state is necessary

[*1] When changing input INCK frequency, care should be taken not to be input pulses whose width are shorter than the High / Low level width in front and behind of the INCK pulse at the frequency change. If the pulses above generate at the frequency change, change INCK frequency during system reset in the state of XCLR = Low, and then perform system clear in the state of XCLR = High following the item of "Power on sequence" in the section of "Power on / off sequence". Execute initial setting again because the register settings become default state after system clear.

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16.11. Other Function

This sensor has the function as below. About detail, refer to each application note.

- Multi Frame Set Output mode (2 / 4 frame)
- Multi Exposure Trigger mode
- Multi Frame ROI (Multi Exposure + ROI) mode
- Low Power Consumption (Low power consumption function is always ON)
- Thermometer
- Gradation Compression
- Dual Trigger
- Self Trigger
- Short Interval Shutter function
- Pattern Generator (Refer to Support Package)

16.12. Extension Function

Use these functions after enough checks and evaluation.

- Black Level Auto Adjust Off
- Short Exposure Mode

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17. Power-on and Power-off Sequence

17.1. Power-on sequence

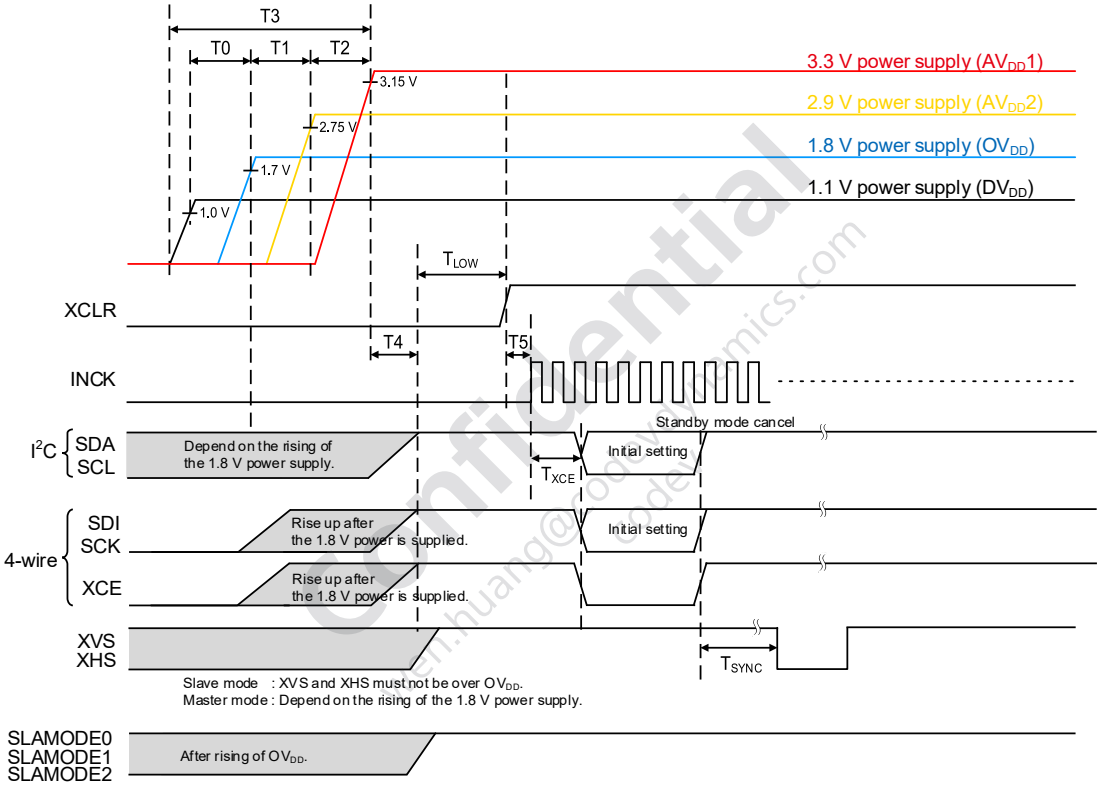
Follow the sequence below to turn on the power supplies.

(1) Turn on the power supplies so that the power supplies rise in order of 1.1 V power supply (DV_{DD}) →1.8 V power supply (OV_{DD}) → 2.9 V power supply ($AV_{DD}2$) → 3.3 V power supply ($AV_{DD}1$). In addition, all power supplies should finish rising within 200 ms. Each digital input terminal (INCK, XCE, SCK, SDI, XCLR, XMASTER, XTRIG1, XTRIG2, SLAMODE0, SLAMODE1, SLAMODE2 and XVS, XHS) set 0V or Hi – Z.

(2) The register values are undefined immediately after power-on, so the system must be cleared. Hold XCLR at Low level for 500 ns or more after all the power supplies have finished rising. (The register values after a system clear are the default values.)
In addition, hold XCE to High level during this period. Rise XCE after 1.8 V power supply (OV_{DD}), so hold XCE at High level until INCK is input.

(3) Start the input of INCK after turning the level of XCLR into the high.

(4) Make sensor settings by register communication after stabilizing the master clock (INCK).



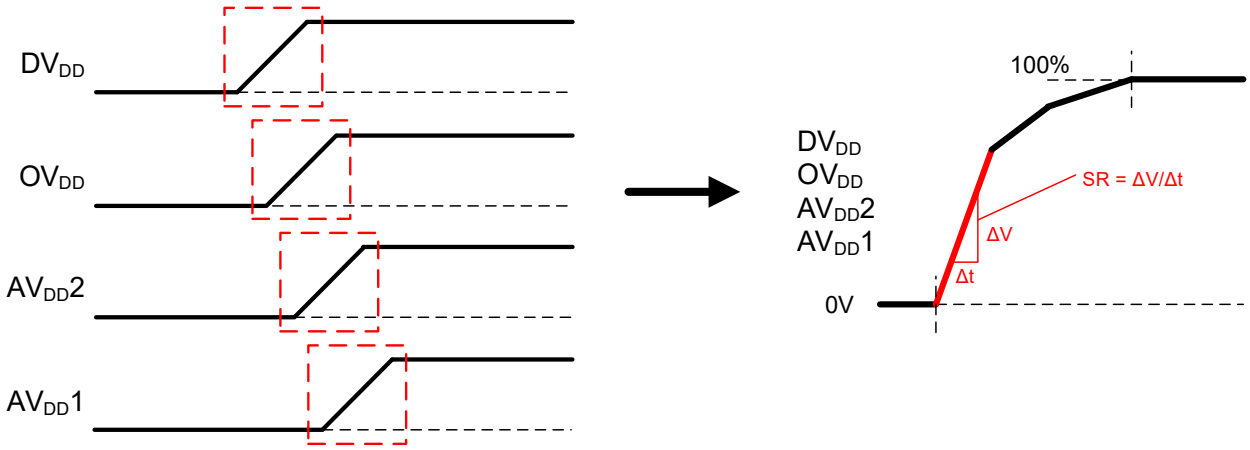
Power-on Sequence

Item	Symbol	Min.	Max.	Unit
1.1 V power supply rising to 1.8 V power supply rising	T0	0	–	ms
1.8 V power supply rising to 2.9 V power supply rising	T1	0	–	ms
2.9 V power supply rising to 3.3 V power supply rising	T2	0	–	ms
Rising time of all power supplies	T3	–	200	ms
All power supplies rising to XCE High	T4	0	–	ms
XCE High to Clear OFF	T_{LOW}	0.5	–	µs
Clear OFF to INCK rising	T5	1	–	µs
INCK rising to Communication start	T_{XCE}	20	–	µs
Standby OFF (communication) to External input XHS, XVS (slave mode only)	T_{SYNC}	6.8	–	ms

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17.2. Slew Rate Limitation of Power-on Sequence

Conform the slew rate limitation shown below when power supply change 0 V to each voltage (0 % to 100 %) in power-on sequence.

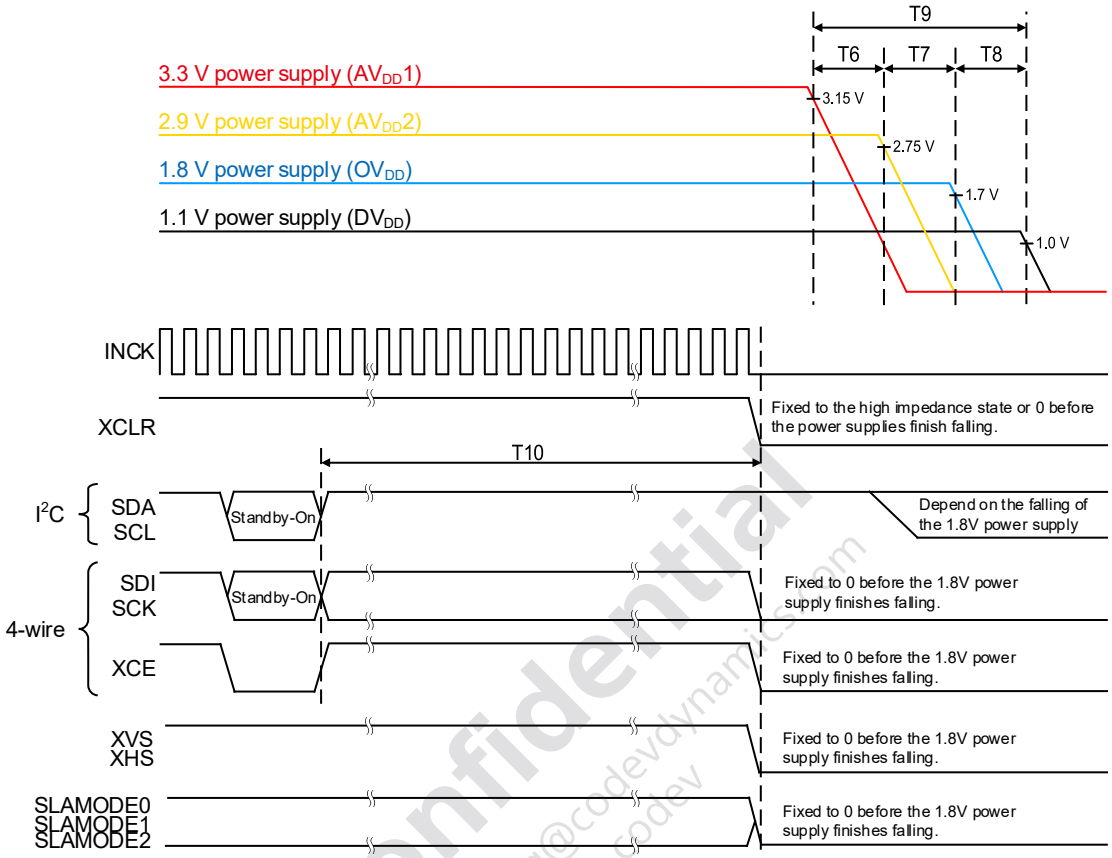


Item	Symbol	Power supply	Min.	Max.	Unit	Remarks
Slew rate	SR	DV_{DD} (1.1 V)	—	25	mV/μs	
		OV_{DD} (1.8 V)	—	25	mV/μs	
		$AV_{DD}2$ (2.9 V)	—	25	mV/μs	
		$AV_{DD}1$ (3.3 V)	—	25	mV/μs	

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17.3. Power-off Sequence

Turn off the power supplies so that the power supplies fall in order of 3.3 V power supply ($AV_{DD}1$) → 2.9 V power supply ($AV_{DD}2$) → 1.8 V power supply (OV_{DD}) → 1.1 V power supply (DV_{DD}). In addition, all power supplies should finish falling within 200 ms.
Set each digital input pin (INCK, XCE, SCK, SDI, XCLR, XMASTER, XTRIG1, XTRIG2, SLAMODE0, SLAMODE1, SLAMODE2, XVS, XHS) to 0 V or high impedance before the 1.8 V power supply (OV_{DD}) falls.



Power-off Sequence

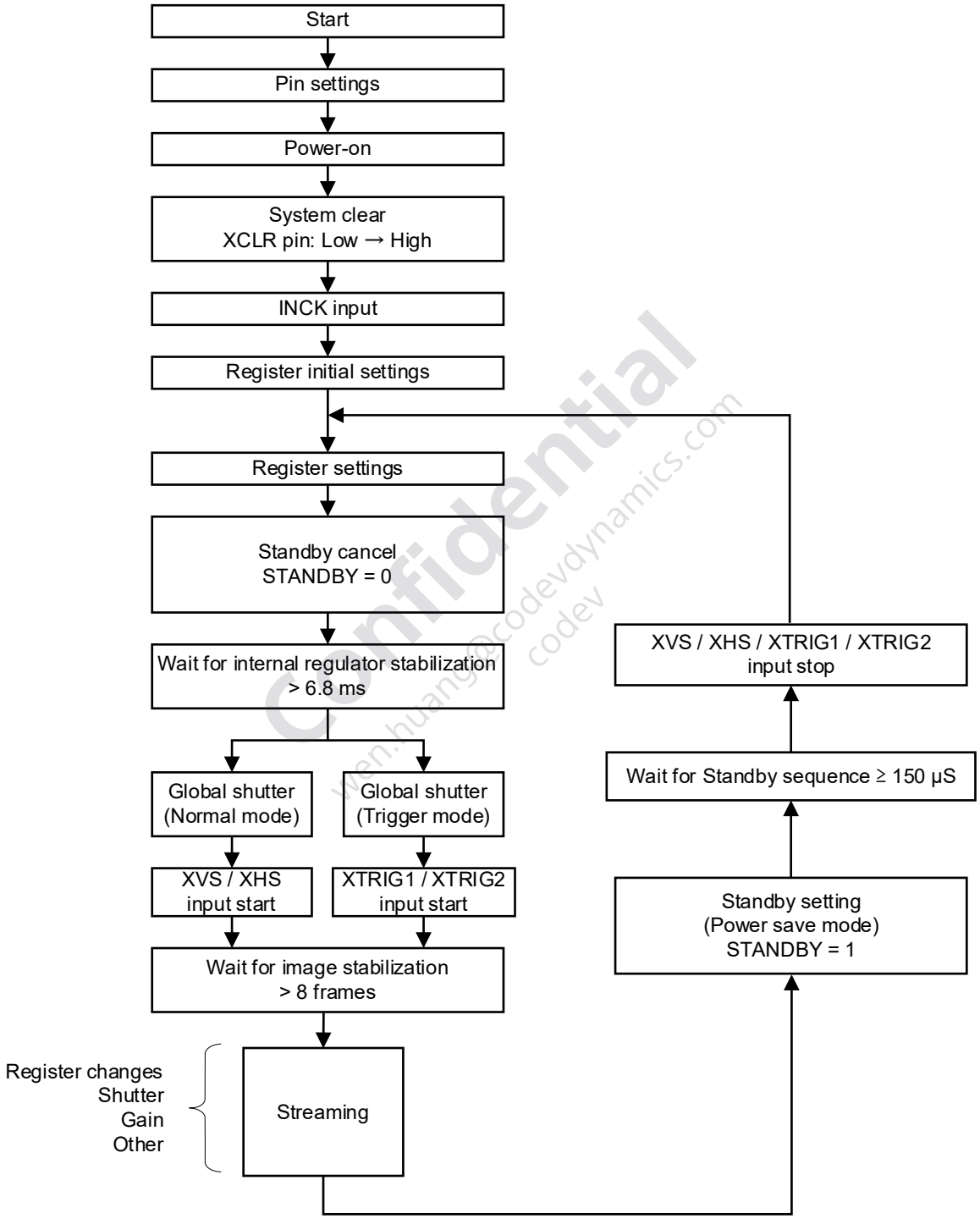
Item	Symbol	Min.	Max.	Unit
3.3 V power supply fall to 2.9 V power supply fall	T6	0	–	ms
2.9 V power supply fall to 1.8 V power supply fall	T7	0	–	ms
1.8 V power supply fall to 1.1 V power supply fall	T8	0	–	ms
Fall time of all power supplies	T9	–	200	ms
“Standby-On” to “Set 0 V or Hi-z each digital input pin”	T10	150	–	µs

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18. Sensor Setting Flow

18.1. Setting Flow in Sensor Slave Mode

The figure below shows the operating flow in sensor slave mode.
For details from “Power-on” to “System clear”, see section “Power-on sequence”. For details from “Standby cancel” to “Wait for image stabilization”, see section “Standby mode”. “Standby setting (Power save mode)” can be made by setting the STANDBY register to “1” during “Streaming”.

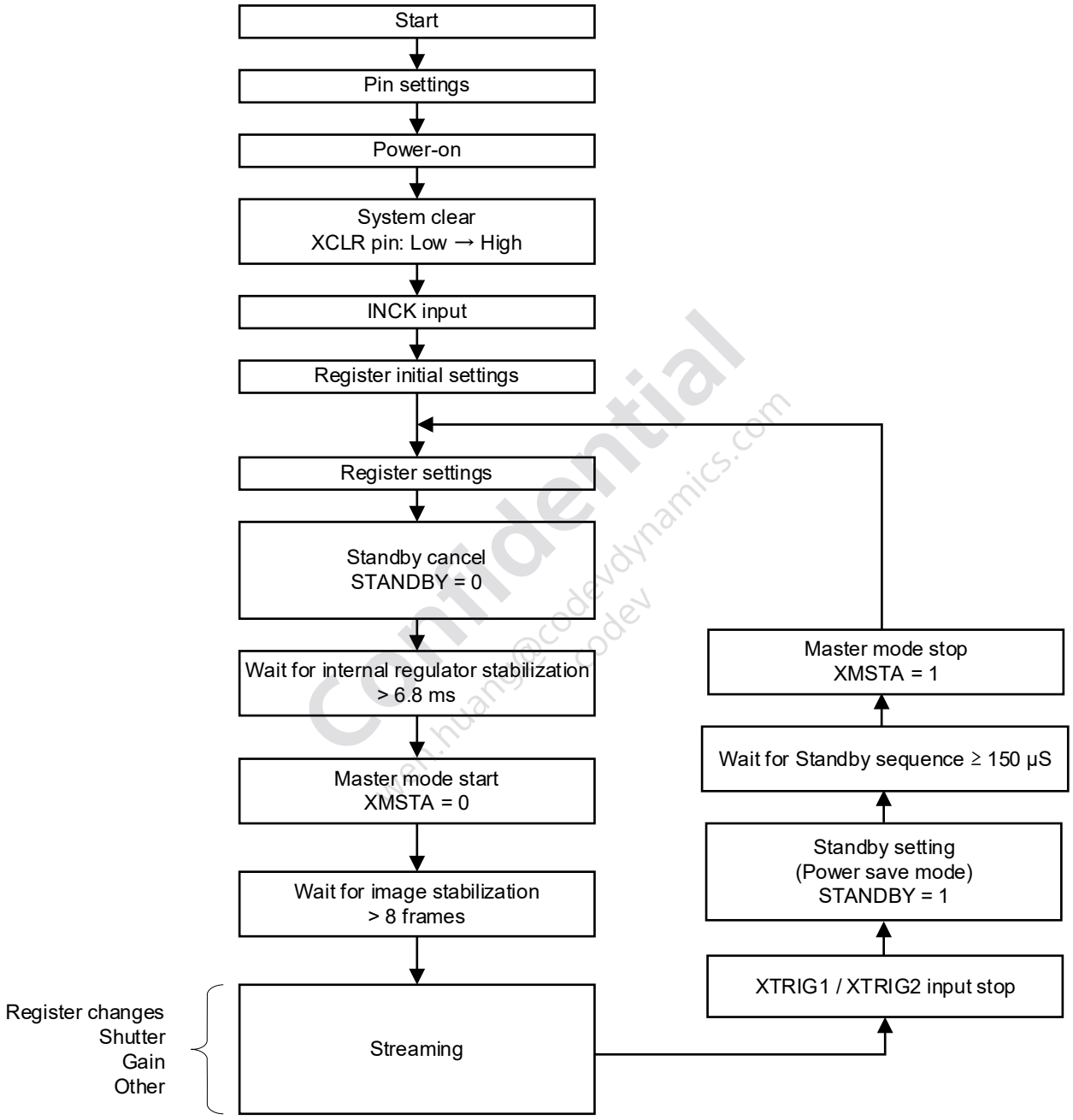


Sensor Setting Flow (Sensor Slave Mode)

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18.2. Setting Flow in Sensor Master Mode

The figure below shows the operating flow in sensor master mode.
For details from “Power-on” to “System clear”, see section “Power-on sequence”. For details from “Standby cancel” to “Wait for image stabilization”, see section “Standby mode”. In master mode, “Master mode start” by setting the master mode start register XMSTA to “0” after “Wait for internal regulator stabilization”.
“Standby setting (Power save mode)” can be made by setting the STANDBY register to “1” during “Streaming”. At this time, set “Master mode stop” by setting XMSTA to “1”.

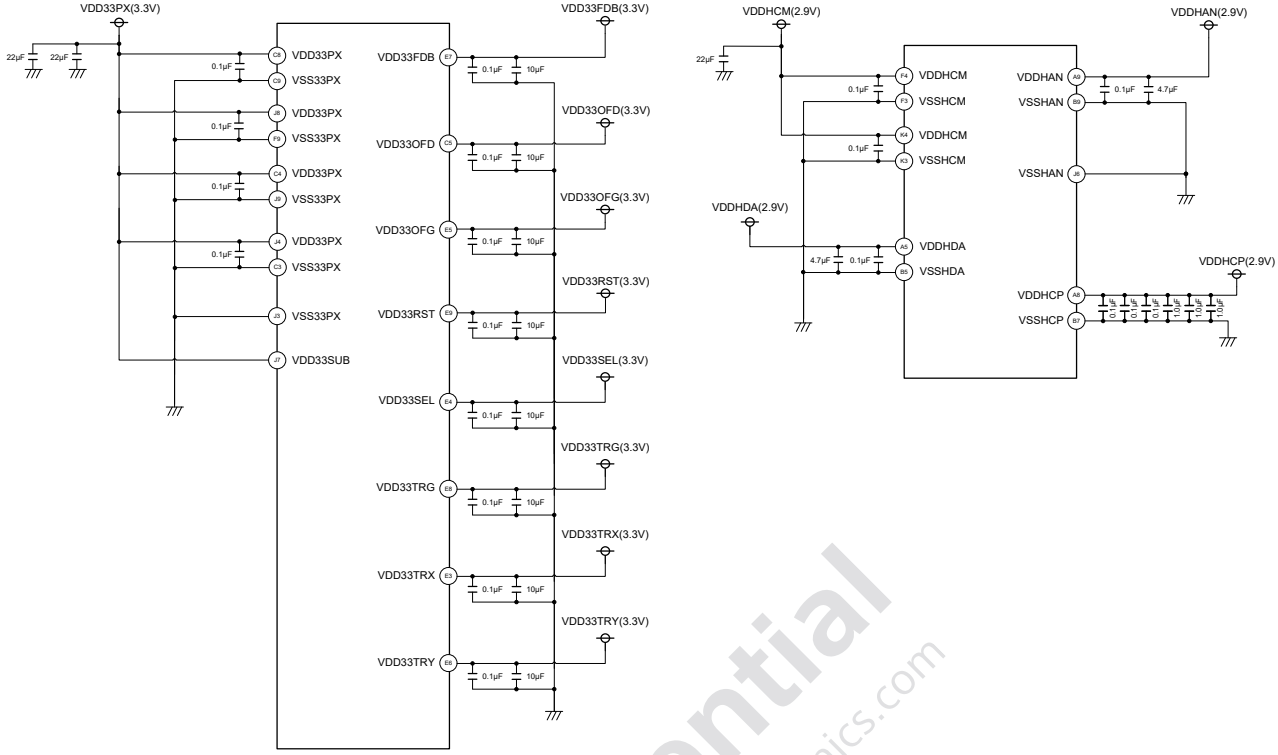


Sensor Setting Flow (Sensor Master Mode)

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19. Peripheral Circuit

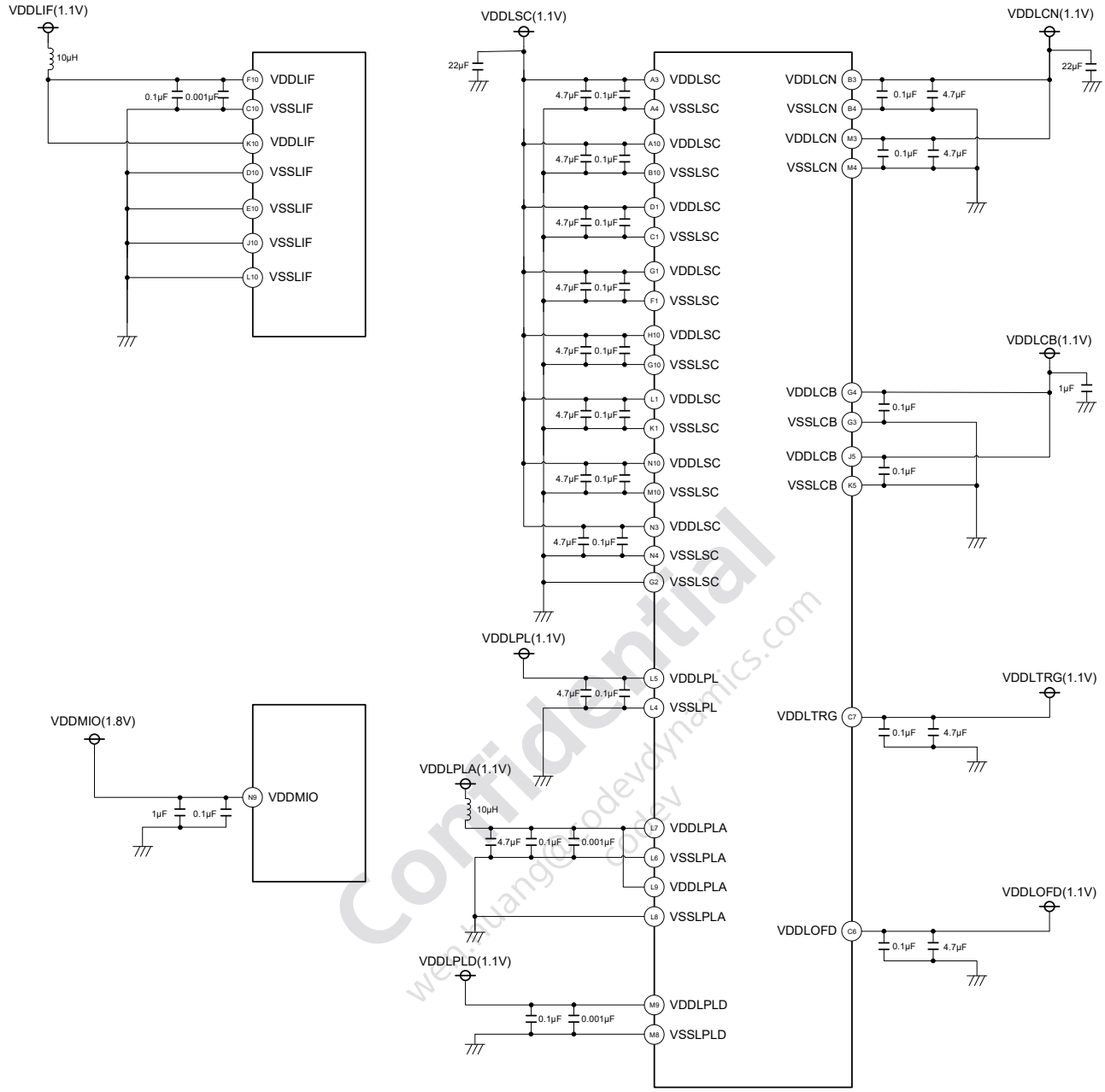
19.1. Analog Power Pins



Application circuits shown are typical examples illustrating the operation of the devices. Sony Semiconductor Solutions Corporation cannot assume responsibility for any problems arising out of the use of these circuits or for any infringement of third party and other right due to same.

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19.2. Digital Power Pins

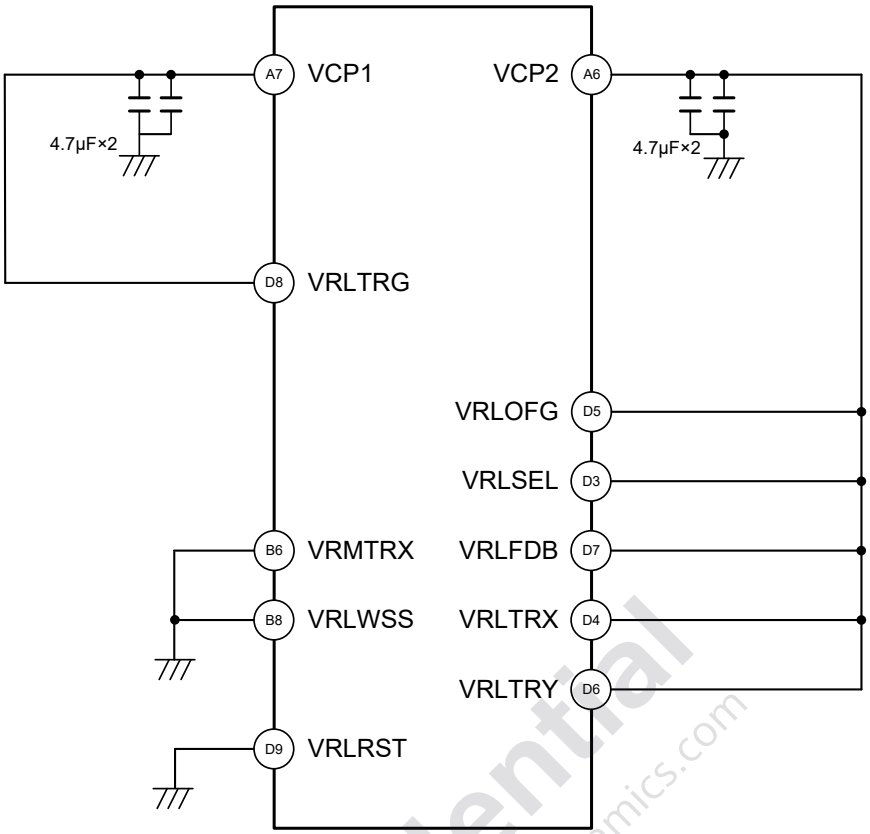


Pin G4, J5, C6, C7 are analog power pins. But these pins can be connected to the digital power pins.
So, it describes on this page. These pins can be separated from the digital power pins.

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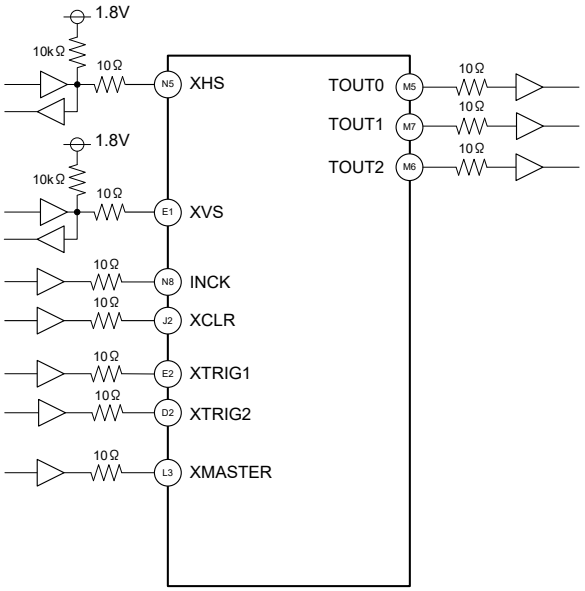
19.3. Analog Other Pins

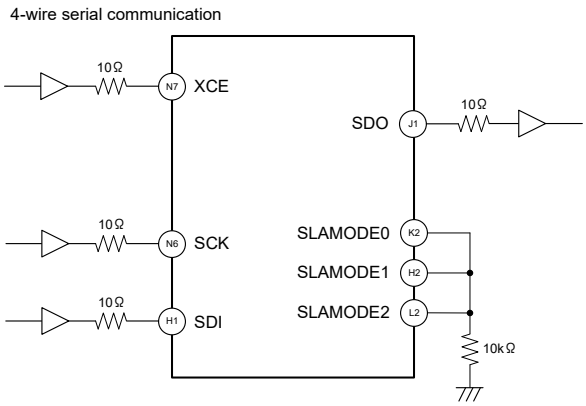


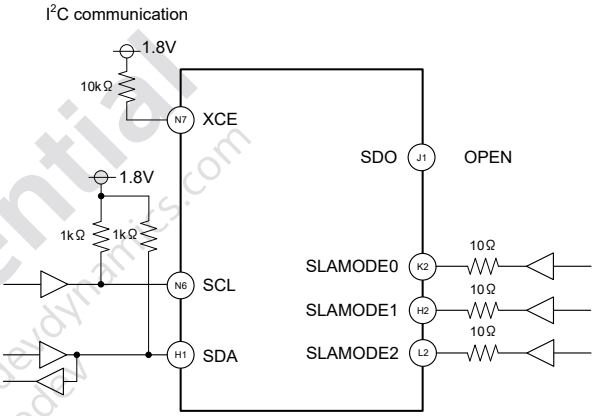
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19.4. Digital I/O Pins



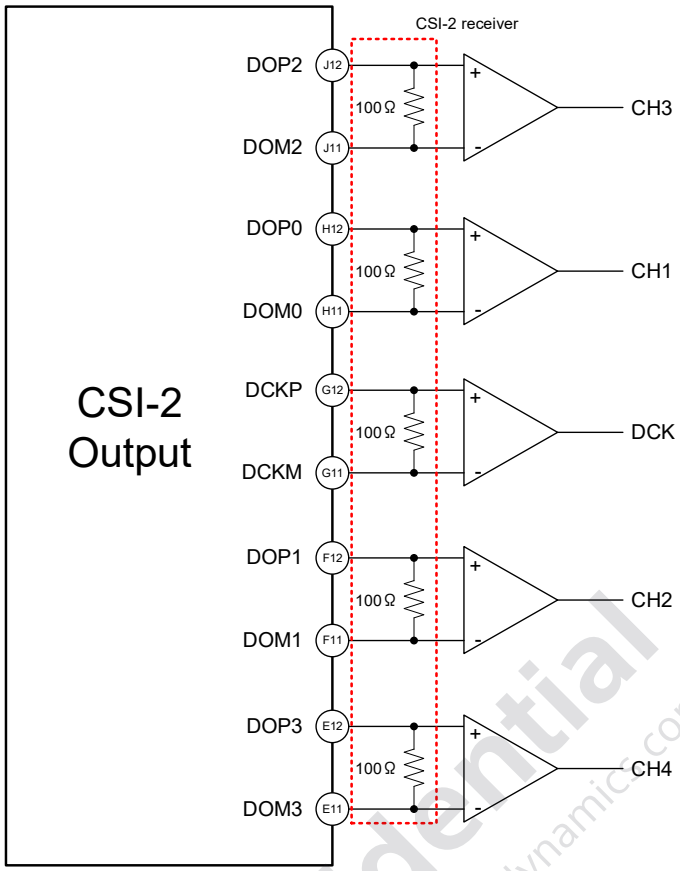




Application circuits shown are typical examples illustrating the operation of the devices. Sony Semiconductor Solutions Corporation cannot assume responsibility for any problems arising out of the use of these circuits or for any infringement of third party and other right due to same.

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19.5. Output pins

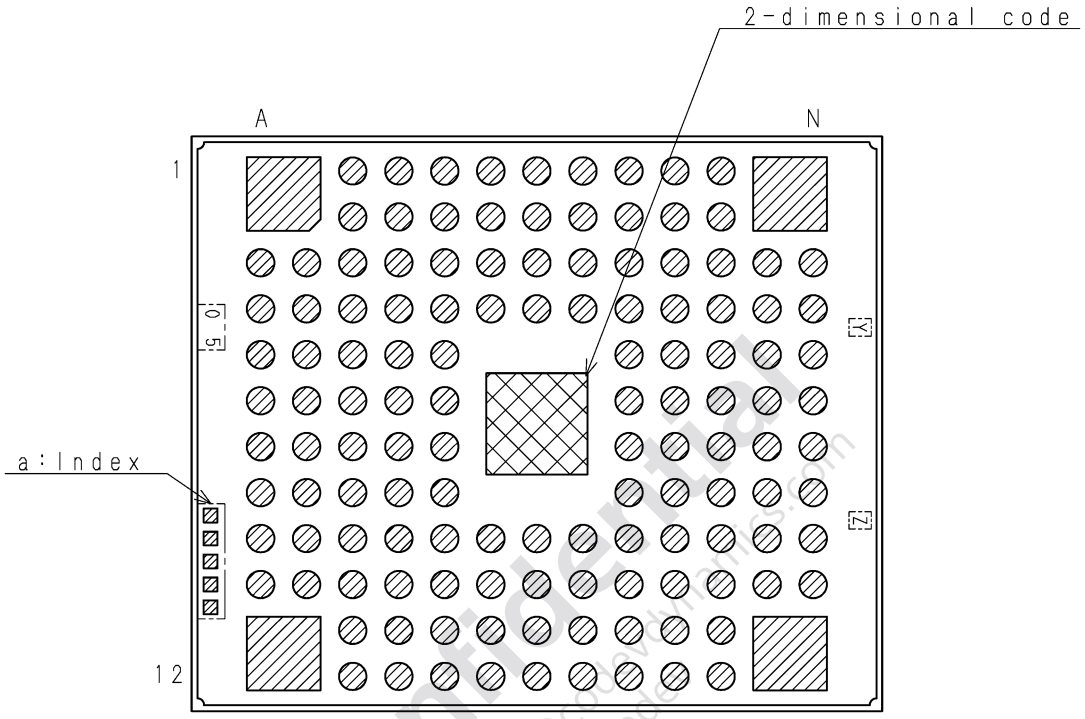


The MIPI IF terminating resistor is not used in LP mode. (Required in HS mode)
Therefore, control the terminating resistor on the receiving side.

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Application circuits shown are typical examples illustrating the operation of the devices. Sony Semiconductor Solutions Corporation cannot assume responsibility for any problems arising out of the use of these circuits or for any infringement of third party and other right due to same.

20. Marking



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Note : One character of alphabet or number shall be placed from Y to Z part.
(Plating option)

As for part "a", up to 5 indexes are arranged.
(Plating option)

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21. Notes On Handling

1. Static charge prevention

Image sensors are easily damaged by static discharge. Before handling be sure to take the following protective measures.

(1) Either handle bare handed or use non-chargeable gloves, clothes or material.
Also use conductive shoes.
(2) Use a wrist strap when handling directly.
(3) Install grounded conductive mats on the floor and working table to prevent the generation of static electricity.
(4) Ionized air is recommended for discharge when handling image sensors.
(5) For the shipment of mounted boards, use boxes treated for the prevention of static charges.

2. Protection from dust and dirt

Image sensors are packed and delivered with care taken to protect the element glass surfaces from harmful dust and dirt. Clean glass surfaces with the following operations as required before use.

(1) Perform all lens assembly and other work in a clean environment (class 1000 or less).
(2) Do not touch the glass surface with hand and make any object contact with it.
If dust or other is stuck to a glass surface, blow it off with an air blower.
(For dust stuck through static electricity, ionized air is recommended.)
(3) Clean with a cotton swab with ethyl alcohol if grease stained. Be careful not to scratch the glass.
(4) Keep in a dedicated case to protect from dust and dirt. To prevent dew condensation, preheat or precool when moving to a room with great temperature differences.
(5) When a protective tape is applied before shipping, remove the tape applied for electrostatic protection just before use. Do not reuse the tape.

3. Installing (attaching)

(1) If a load is applied to the entire surface by a hard component, bending stress may be generated and the package may fracture, etc., depending on the flatness of the bottom of the package. Therefore, for installation, use either an elastic load, such as a spring plate, or an adhesive.
(2) The adhesive may cause the marking on the rear surface to disappear.
(3) If metal, etc., clash or rub against the package surface, the package may chip or fragment and generate dust.
(4) Acrylate anaerobic adhesives are generally used to attach this product. In addition, cyanoacrylate instantaneous adhesives are sometimes used jointly with acrylate anaerobic adhesives to hold the product in place until the adhesive completely hardens. (Reference)
(5) Note that the sensor may be damaged when using ultraviolet ray and infrared laser for mounting it.

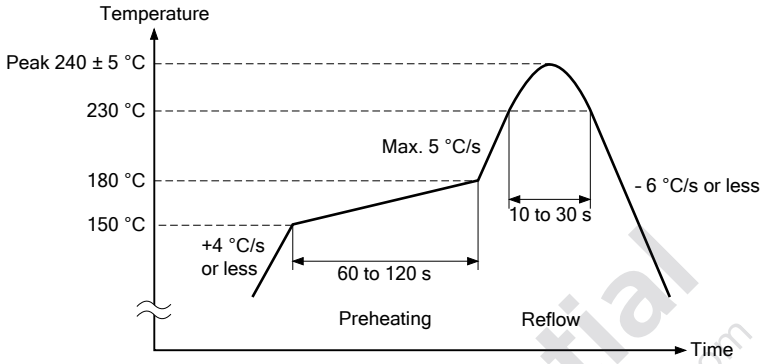
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4. Reflow soldering conditions

The following items should be observed for reflow soldering.

(1) Recommended temperature profile for reflow soldering

Control item	Profile (at part side surface)
1. Preheating	150 to 180 °C 60 to 120 s
2. Temperature up (down)	+4 °C/s or less (- 6 °C/s or less)
3. Reflow temperature	Over 230 °C 10 to 30 s Max. 5 °C/s
4. Peak temperature	Max. 240 ± 5 °C



(2) Reflow conditions

(a) Make sure the temperature of the upper surface of the seal glass resin adhesive portion of the package does not exceed 245 °C.

(b) Perform the reflow soldering only one time.

(c) Finish reflow soldering within 72 h after unsealing the degassed packing.
Store the products under the condition of temperature of 30 °C or less and humidity of 70 % RH or less after unsealing the package.

(d) Perform re-baking only one time under the condition at 125 °C for 24 h.

(e) Note that condensation on glass or discoloration on resin interfaces may occur if the actual temperature and time exceed the conditions mentioned above.

(3) Others

(a) Carry out evaluation for the solder joint reliability in your company.

(b) After the reflow, the paste residue of protective tape may remain around the seal glass.
(The paste residue of protective tape should be ignored except remarkable one.)

(c) Note that X-ray inspection may damage characteristics of the sensor.

5. Others

(1) Do not expose to strong light (sun rays) for long periods, as the color filters of color devices will be discolored.

(2) Exposure to high temperature or humidity will affect the characteristics. Accordingly avoid storage or use in such conditions.

(3) This product is precision optical parts, so care should be taken not to apply excessive mechanical shocks or force.

(4) Note that imaging characteristics of the sensor may be affected when approaching strong electromagnetic wave or magnetic field during operation.

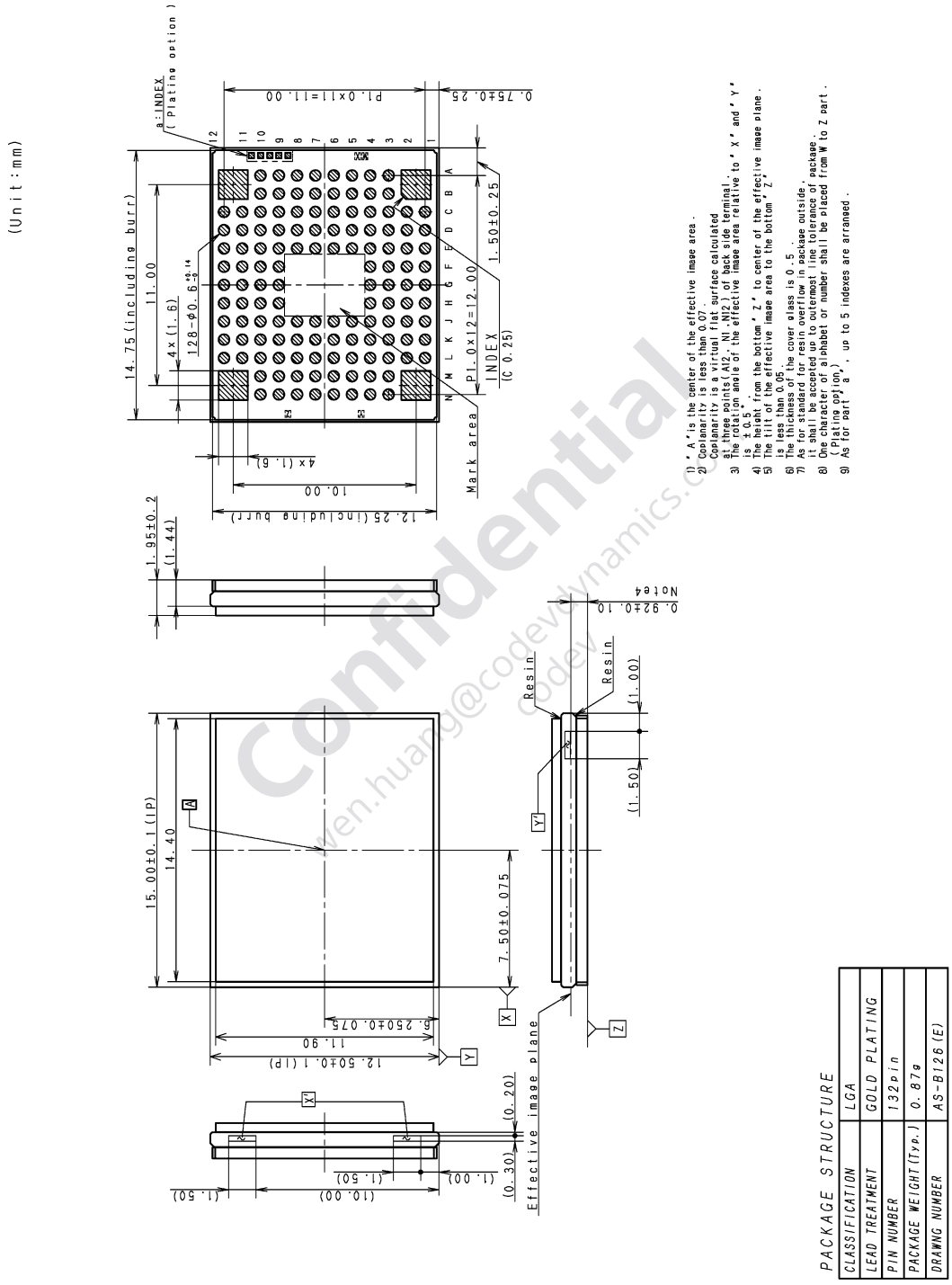
(5) Note that image may be affected by the light leaked to optical black when using an infrared cut filter that has transparency in near infrared ray area during shooting subjects with high luminance.

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22. Package Outline

(Unit: mm)



1) 'A' is the center of the effective image area.
2) Coplanarity is less than 0.07.
Coplanarity is a virtual flat surface calculated at three points(A12, N1, N12) of back side terminal.
3) The rotation angle of the effective image area relative to 'X' and 'Y' is ±0.5°.
4) The height from the bottom 'Z' to center of the effective image plane.
5) The tilt of the effective image area to the bottom 'Z'
is less than 0.05.
6) The thickness of the cover glass is 0.5.
7) As for standard for resin overflow in package outside, it shall be accepted up to outermost line tolerance of package.
8) One character of alphabet or number shall be placed from W to Z part.
(Plating option)
9) As for part 'a', up to 5 indexes are arranged.

PACKAGE STRUCTURE

CLASSIFICATION	LGA
LEAD TREATMENT	GOLD PLATING
PIN NUMBER	132pin
PACKAGE WEIGHT(Typ.)	0.87g
DRAWING NUMBER	AS-B126(E)

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23. List of Trademark Logos and Definition Statements

Pregius S

* Pregius S is a registered trademark or trademark of Sony Group Corporation or its affiliates. Pregius S is a global shutter sensor technology for active pixel-type CMOS image sensors. By stacking the signal processing on the back illuminated type CMOS Image Sensor it realizes small chip size and high sensitivity, whilst using the high picture quality global shutter pixel technology of Pregius.

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